

Embedded framework for low-cost pavement condition evaluation using microcontroller and single-board computer platforms

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ARTICLE INFO

Keywords:

Pavement evaluation
Embedded systems
Low-cost sensors
Microcontrollers
Single-board computers
Edge computing
Raspberry Pi

ABSTRACT

Traditional automated pavement evaluation methods, while effective, are often prohibitively expensive for transportation agencies with limited budgets. This paper presents a framework for developing low-cost embedded-based systems for pavement condition evaluation using microcontroller and single-board computer platforms. The paper focuses on providing guidance to transportation agencies for creating customized pavement monitoring solutions that are low-cost and flexible. A comprehensive set of criteria for component selection and system integration are discussed from a civil engineering perspective, emphasizing practical considerations. To demonstrate the framework's application, a case study featuring the development of a multi-sensor device integrating various sensors with a single-board computer platform is presented. The proof-of-concept implementation validates the framework's effectiveness for creating tailored pavement monitoring systems. Preliminary results confirm the system's data collection capabilities across multiple pavement condition indicators. The methodology, selection criteria, and implementation considerations provide a foundation for transportation agencies to develop affordable, fit-for-purpose pavement evaluation solutions.

1. Introduction

Roadway pavement is considered the backbone of a nation's economy, providing critical transportation networks to facilitate economic growth [1,2]. However, investment in road construction and maintenance is expensive. Due to budget constraints, needed repair activities are often deferred [2–6]. Delaying maintenance coupled with increased traffic loads and environmental factors can cause pavements to deteriorate structurally and functionally well before their intended design life. This accelerated deterioration exponentially increases future repair costs. Further postponing maintenance beyond the pavement's design service life will likely result in complete failure, requiring full reconstruction - a highly uneconomical outcome [2]. To address these challenges, a substantial amount of capital is allocated annually for pavement condition monitoring and maintenance activities. For countries like the United States, possessing extensive pavement networks spanning 2.8 million miles [7], the costs associated with maintenance, rehabilitation and reconstruction across all public roads can be remarkably high, with estimates surpassing \$60 billion in 2022 [8]. Additionally, monitoring expenses for asset management of such vast infrastructure are significantly high given the current reliance on costly

automated pavement data collection vehicles [9].

Transportation agencies have historically developed pavement management systems (PMS) to mitigate deterioration and make informed maintenance decisions [3,10]. PMS comprise tools and systematic processes for planning, analyzing, and timely maintenance using appropriate materials within budget constraints. The modular PMS components include data collection, performance modeling, decision analysis, maintenance implementation, and monitoring. However, PMS decisions depend heavily on the quality of collected data, analytical techniques, budget availability, and stakeholder influence [3]. Data collection is crucial in PMS as it affects overall pavement efficiency. High quality data enables advanced analysis and informed decision-making. Pavement condition data guides infrastructure planning and maintenance decisions for transportation agencies. Parameters such as roughness, cracking, rutting and macrotexture are routinely measured to assess functional pavement performance [4]. However, collecting these data through pavement data vans equipped with specialized sensors can cost over \$106 per kilometer (\$170 per mile) [11]. Labor requirements are also intensive for comprehensive network surveys that may need to be repeated every 1–2 years [12].

State agencies have increasingly used automated vehicles with

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<https://doi.org/10.1016/j.autcon.2025.106442>

Received 10 February 2025; Received in revised form 25 July 2025; Accepted 28 July 2025

Available online 31 July 2025

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cameras, lasers and sensors to collect pavement condition data, though these are costly [2,13–15]. Although state transportation agencies own the backbone of the road network in the US, other jurisdictions like counties, municipalities, and townships are also responsible for significant pavement assets. For example, in 2020 almost 3.4 million miles of the roads in the US are owned by non-state highway agencies, but only around 782 thousand miles of roads are owned by state highway agencies [16]. Notably, automated pavement data collection methods are predominantly utilized by state Department of Transportation (DOTs) given the high costs involved. Smaller agencies with limited budgets often struggle to comprehensively evaluate pavement conditions and effectively implement their PMS [17]. Unlike state DOTs, many local transportation agencies lack sufficient technical capability [18]. These factors lead to inefficient maintenance plans and faster deterioration of locally owned roads compared to state highways.

Low-cost pavement evaluation devices can help small/medium agencies overcome budget and technical barriers to effectively assess pavement condition. Small, inexpensive accelerometers, Global Positioning System (GPS) modules, microcontrollers, Single Board Computers (SBCs), and other type of sensors can be combined to create flexible monitoring devices [19]. These systems can be temporarily mounted on any vehicle traveling at normal traffic speeds, enabling frequent data collection across entire networks without major resource commitments or traffic impacts. This paper presents a framework for developing low-cost embedded-based systems for pavement condition evaluation, designed to guide transportation agencies in creating their own monitoring solutions based on their specific requirements. The framework provides comprehensive criteria for component selection and system integration from a civil engineering perspective, emphasizing practical considerations and budget constraints. To demonstrate the framework's application, a case study is presented featuring the development of a multi-sensor device that integrates various sensors with an SBC computer platform for pavement evaluation. This proof-of-concept implementation demonstrates the practical application of the proposed framework for creating customized pavement monitoring solutions. The development methodology, selection criteria, and implementation considerations are presented, along with preliminary results validating the system's data collection capabilities.

2. Background

Pavement condition assessment is fundamental to effective infrastructure management. Evaluation methods have evolved from simple visual inspections to sophisticated automated systems, driven by the need for objective and efficient techniques applicable across large networks. Modern approaches can be categorized by their sensing technologies and implementation platforms, with increasing interest in affordable solutions that expand access to quality pavement data.

2.1. Vibration-based assessment approaches

Vibration-based pavement condition assessment methods have gained considerable attention over recent years due to their cost-effectiveness, efficiency, and adaptability for continuous monitoring. These methods primarily rely on IMUs, including accelerometers and gyroscopes, mounted on vehicles to capture pavement-induced vibrations.

Early research in this field began with smartphone applications using embedded sensors. These applications can collect and analyze data on pavement distresses, such as potholes, cracks, and roughness using the smartphone's accelerometer, gyroscope, GPS, and camera sensors. Several studies have developed and validated various smartphone-based systems. The Microsoft Research team first explored using GPS and accelerometer sensors in smartphones to detect pavement irregularities like potholes [20,21]. Subsequent studies developed smartphone-based systems [22–26] that analyze acceleration, GPS, and video data to

evaluate pavement condition. These systems demonstrated the feasibility and accuracy of smartphone-based pavement condition monitoring as well as benefits of crowdsourcing data.

Recent methodologies primarily involve inertial profiling, in which the collected vibration data is processed through algorithms that correlate vehicle vibrations to standard pavement indices such as the International Roughness Index (IRI) [27]. Machine learning algorithms, including Random Forest, Support Vector Machines (SVM), and Fully Connected Neural Network (FCNN) have enhanced the accuracy of identifying pavement anomalies and classifying pavement conditions from vibration data [28]. For instance, research has demonstrated the effectiveness of combining IMU sensors with GPS receivers to detect, classify, and map potholes based on their size and severity, achieving detection rates exceeding 90 % and classification accuracy above 85 % [29]. Advanced applications have leveraged inertial navigation system (INS) data from vehicles to detect minor pavement damage. By analyzing vehicle attitude changes through accelerometers, gyroscopes, and GPS, models have achieved exceptional accuracy with F1 scores of 99.61 % and precision of 99.33 % in classifying road conditions [30].

The integration of multiple sensing technologies has further improved detection reliability. Crowdsourcing-based multi-sensor fusion approaches have shown that combining accelerometer data with in-vehicle video can improve detection accuracy by approximately 6 % compared to traditional single-sensor methods, effectively addressing the limitations of accelerometer sensitivity to driving speed and image recognition's vulnerability to ambient light conditions [31]. A significant advancement in this field is the development of semi-supervised learning models to address challenges of incomplete data and multi-vehicle data fusion. This approach leverages crowdsourced data from ordinary vehicles, offering extensive coverage at lower costs with relative errors below 10 % compared to traditional assessment methods [32].

Despite their promise, vibration-based pavement assessment approaches face several limitations. Vibration signals are often influenced not only by the road surface but also by vehicle speed, suspension properties, and sensor mounting position [33]. Factors like smartphone configurations, sensor quality, vehicle/driver dynamics, and pavement environment can affect data quality and reliability [34–36]. Sensor noise and spurious vibrations can occlude useful information and hinder correct anomaly detection, presenting ongoing challenges for researchers in this field.

2.2. Vision-based assessment approaches

Vision-based pavement condition assessment methods have rapidly advanced due to significant developments in computer vision (CV), machine learning (ML), and deep learning (DL) technologies. These methods primarily utilize images captured through various platforms and sensors to detect, classify, segment, and quantify pavement distresses [37].

Early vision-based techniques involved basic image processing and feature extraction methods, which gradually evolved to incorporate sophisticated CV techniques. Various approaches for defect detection in concrete and asphalt pavement have been documented [38], highlighting the effectiveness of integrated image analysis techniques for pavement condition assessment. Building on this foundation, Ye et al. explored DL-based structural health monitoring for civil infrastructures, demonstrating how these advanced models can handle complex visual patterns with high accuracy [39].

Recent research has introduced deep convolutional neural networks (CNNs) as powerful tools for pavement distress detection, significantly outperforming traditional image-processing methods. CNN-based models have shown remarkable capability in detecting cracks, potholes, and other pavement anomalies with high precision [40,41]. Zhang et al. developed a Deep CNN model demonstrating superior results compared to conventional ML techniques [42]. Additionally,

advanced object detection frameworks like YOLO (You Only Look Once) have enhanced real-time detection capabilities for practical applications on large road networks [43,44].

Unmanned Aerial Vehicles (UAVs) have emerged as promising platforms for lower-cost pavement monitoring. For example, Zhang used a UAV helicopter equipped with a video camera to detect pavement distresses like wash boarding, aggregate loss and potholes [45]. Dobson et al. mounted a DSLR camera on electric helicopter UAV to survey cracking, potholes, and roughness [46]. Wu et al. and Leonardi et al. utilized off-the-shelf UAVs with built-in cameras to assess cracking, raveling, and potholes [47,48]. These studies indicate that UAVs can provide comprehensive and accurate information on various pavement condition indicators and facilitate efficient pavement management.

Despite these advancements, vision-based approaches face several challenges. Image quality variability due to lighting conditions, camera angle, and environmental factors affects assessment consistency. Additionally, UAV-specific limitations include restrictions on operating zones, flight ceilings, payload weights, meteorological factors, transmission distance limitations, and battery life constraints [49,50].

2.3. Low-cost embedded systems for pavement monitoring

In recent years, low-cost embedded-based sensor systems have emerged as promising alternatives for automated and continuous pavement monitoring. Arbabpour Bidgoli et al. developed a road roughness monitoring system utilizing an Arduino microcontroller and wheel-mounted accelerometers [51]. Field tests on a calibration test section demonstrated the system could measure International Roughness Index (IRI) with high accuracy compared to a reference profiler, with a coefficient of determination (R^2) of 0.87 and average error below 20 %. Similarly, Mirtabar et al. developed a crowdsourcing approach for IRI measurement using an affordable sensor package integrated by an Arduino Mega microcontroller [52]. Accelerometer and GPS data logged on-board was post-processed into IRI values using ProVAL software, achieving adequate accuracy versus reference data with 15 % average error. Furthermore, Loprencipe et al. proposed an inertial sensor-based system using a low-cost Inertial Measurement Unit (IMU), GPS module, and Raspberry Pi Zero W SBC embedded in a vehicle [53]. Promising correlations were shown between the system readings and typical pavement condition indices like IRI and ride number ($R^2 = 0.83\text{--}0.90$). They demonstrated the proposed sensor as a reliable and easy-to-install method for monitoring pavement conditions in urban networks. Similarly, Bruno et al. utilized accelerometers mounted on a car and bicycle, along with a Raspberry Pi, GPS receiver, and other sensors to monitor the riding quality of stone pavements [54]. Regression models were also developed to estimate conventional pavement performance indicators from the vibration values measured across the entire network. Field testing validated the reliability and usefulness of this innovative approach for assessing and preserving historically significant stone road surfaces.

Diallo Diene and Ndiaye designed a multichannel analysis of surface waves (MASW) acquisition device for pavement layer investigation [55]. The authors developed a seismograph using an Arduino Due microcontroller connected to a Raspberry Pi 4 SBC, which acts as a real-time data visualization and processing interface. The receivers consist of six 4.5 Hz GD geophones connected by a graduated tape to acquire data in land-streamer mode. The recording is triggered by a KY-038 sound sensor. The seismic data are processed using the Geopsy open-source software to extract and invert the dispersion curves of surface waves. The device was tested on four different sites and the obtained seismic velocities were used to deduce the elastic properties of the pavement layers. The results were consistent with the expected values and showed the feasibility and effectiveness of the device for pavement investigation. Meanwhile, Mohammed et al. explored using piezoelectric (PZT) sensors integrated with a Raspberry Pi platform to evaluate flexible pavement structural strength through non-destructive testing

[56]. By applying oscillating waves from the microcontroller to actuator PZT sensors and reading responses with receiver PZT sensors, pavement tests were conducted on both traditional lab samples and in-situ using the IoT module approach. Results demonstrated the feasibility of the IoT system for pavement health monitoring applications.

Several studies deployed ML and CV techniques on microcontrollers and SBCs for real-time pavement characterization from sensor readings. Wang et al. classified road surface types using motion sensor data and classifiers embedded on a Raspberry Pi 3B+, attaining over 95 % accuracy for detection [57]. Garcillanosa et al. developed an affordable smart pothole detection device using a Raspberry Pi platform [58]. The key components included a camera module to capture road surface images, a GPS module to obtain location data, and a Wi-Fi device to send reports. Python and OpenCV were utilized on the Raspberry Pi to perform real-time image processing and pothole detection via a trained classifier model. The researchers reported an average pothole detection accuracy of 93.72 % based on test data, indicating the viability of their vision-based approach. The prior work on low-cost embedded-based sensor pavement monitoring systems highlights several key advantages over traditional monitoring approaches. As demonstrated in the previous studies, these systems enable continuous data collection in real-time at a relatively low cost by leveraging affordable commercial off-the-shelf (COTS) components. Their compact size allows flexible deployment on existing vehicles to gather frequent network-level monitoring at broader scales. The automated analysis on these platforms eliminates time-intensive manual surveys. Onboard processing avoids lengthy lab testing. In summary, the integration of microcontrollers and SBCs, sensors, and AI enables scalable, intelligent, and low-cost pavement assessment not achievable through traditional methods. These advantages make low-cost embedded systems a promising platform for the next generation of pavement monitoring, as validated across several application examples thus far. Table 1 provides a summary of the low-cost embedded-based systems discussed in this section.

3. Microcontroller and single board computer platforms

The development of effective low-cost pavement evaluation systems begins with selecting appropriate computing platforms that can integrate sensors, process data, and operate reliably under field conditions. This section presents a framework for choosing between microcontroller and single-board computer platforms based on specific application requirements and constraints.

3.1. What are microcontrollers and single board computers

Embedded systems for pavement evaluation can be built using either microcontrollers or SBCs, each offering distinct capabilities and trade-offs. Microcontrollers are integrated circuits that can perform various tasks by combining processor, memory, and input/output (I/O) interfaces on a single chip. They are widely used in embedded systems and Internet of Things (IoT) applications that require both processing functionality and agile, responsive interaction with other components. Microcontrollers can be programmed to control, process, and store information in different ways, depending on the user's needs and the specific application [61]. While SBCs are complete computers built on a single circuit board that incorporate processor, memory, storage, and various I/O interfaces [62–65]. Unlike microcontrollers, SBCs run full operating systems (typically Linux-based) and offer significantly more processing power and memory resources. They are designed for applications requiring complex computations, multimedia capabilities, and sophisticated software stacks. SBCs can handle multiple tasks simultaneously and support high-level programming languages and frameworks, making them particularly suitable for applications involving data analysis, machine learning, and advanced signal processing. Table 2 shows examples and comparison of popular types of microcontrollers and SBCs.

Table 1

Summary of Low-Cost Devices for Pavement Condition Evaluation.

Reference	Platform	Sensors Used	Additional Components	Purpose	Key Results
Arbabpour Bidgoli et al. [51]	Arduino Due R3	ADXL345 accelerometer, GSM-SIM908 module	Laptop	Evaluating pavement roughness	$R^2 = 0.87$, errors < 20 % compared to reference device
Diallo Diene and Ndiaye [55]	Arduino Due and Raspberry Pi 4	KY037 sound sensor, 4.5 Hz GD geophone	LCD screen	Pavement layers investigation	Elastic properties, mechanical quality, data integration
Mohammed et al. [56]	Raspberry Pi 4B	Piezoelectric sensors	ADS115 analog to digital converter, MCP4725 digital to analog converter	Evaluate the strength of flexible pavements	High correlation with traditional methods
Mirtabar et al. [52]	Arduino Mega 2560 r3	Sim808 GPS, MPU9255 accelerometer	Data logging shield for Arduino	Evaluating pavement roughness	$R^2 = 0.82$, errors = 15 % compared to reference device
Wang et al. [57]	Raspberry Pi 3B+	MPU9250 accelerometer and gyroscope	–	Road surface classification	>95 % classification accuracy
Garcillanosa et al. [58]	Raspberry Pi	GPS module, Camera module	Portable Wi-Fi device, LEDs	Automatic detection and reporting of potholes	>93 % detection accuracy
Loprencipe et al. [53]	Raspberry Pi Zero W	MPU9250 accelerometer and gyroscope, GPS module	–	Evaluating pavement roughness	High correlation ($R^2 = 0.83$ – 0.90) with typical pavement indices
Bruno et al. [54]	Raspberry Pi	MPU9250 accelerometer and gyroscope, NEO-6 M GNSS	–	Monitoring stone pavement conditions	High correlation ($R^2 = 0.96$ and 0.74) with PCI and IRI
Tello-Cifuentes et al. [59]	Raspberry Pi Zero	ZED 2I stereo camera, G28U7FTTL GPS module	Laptop	Pothole damage detection	3D images of pavement damage had a relative volume measurement error of 0.80 %
Fiorita et al. [60]	ESP32 32-bit Xtensa LX6 microcontroller	KX132 accelerometer, SHT31 temperature and humidity sensor, Hydroprobe Click moisture content sensor, SM3 LVDT displacement sensor	LoRA SX1276 transceiver	Continuous pavement health monitoring	Reliable and cost-effective for urban roads

Table 2

A Comparison of popular microcontrollers and SBCs.

Platform Type	Examples	Advantages	Disadvantages
Microcontrollers	Arduino Uno Rev3, ESP32-WROOM, Raspberry Pi Pico, STM32F746G-DISCO	Low power consumption, Real-time control, Simple programming, Low cost [66–68]	Limited processing power, Basic development environment, Minimal OS capabilities [69–71]
	Raspberry Pi 4B, Orange Pi 5, NVIDIA Jetson Orin Nano	High processing power, Full OS support, Advanced connectivity, Multimedia capabilities [65,72–75]	Higher power consumption, More complex setup, Higher cost [66,76]

3.2. Criteria for selecting microcontrollers and single board computers

The selection of an appropriate platform for pavement evaluation systems requires careful consideration of multiple criteria that impact the system's performance, cost-effectiveness, and long-term maintainability. These criteria include the purpose and complexity of the intended application, cost considerations, I/O capabilities, real-time processing requirements, and available community support resources. Each criterion plays a crucial role in determining whether a microcontroller, SBC-based solution, or both would be more suitable for specific pavement monitoring applications. Fig. 1 presents an overview of these selection criteria and their key considerations. The following sections discuss each criterion in detail.

3.2.1. Purpose and complexity

The selection between microcontroller and SBC is a critical decision in the development of a low-cost device for pavement evaluation. This choice should be driven by the specific application requirements and the

complexity of tasks the device is expected to perform. Microcontrollers excel at simple, dedicated tasks with straightforward I/O operations, while SBCs are designed for complex, multi-faceted operations requiring significant processing power. This fundamental difference drives the initial selection process when designing pavement monitoring systems.

Microcontrollers are optimized for direct hardware control and real-time response, making them effective for data collection and monitoring tasks in pavement evaluation systems. Their architecture allows efficient handling of multiple sensors simultaneously – such as accelerometers, gyroscopes, and GPS modules - while maintaining precise timing control in data acquisition [66–68,77]. For instance, in roughness measurements, microcontrollers can manage continuous data collection from these sensors and execute predefined algorithms to calculate riding quality indices. Their deterministic behavior and operation without an operating system overhead ensure reliable and consistent performance. However, microcontrollers have limitations in processing power compared to SBCs. While they excel at real-time data acquisition and basic calculations, they are not suited for complex data analysis [69–71]. Applications requiring advanced processing capabilities, such as statistical analysis, pattern recognition in pavement distress detection, or integration with sophisticated visualization tools, necessitate the use of an SBC. This additional processing capability of SBCs comes with trade-offs of higher power consumption and increased system complexity. Table 3 shows example options for common types of microcontrollers along with their specifications.

SBCs offer increased processing power and flexibility compared to traditional microcontrollers. Their ability to run full-fledged operating systems allows SBCs to multitask and integrate a wider range of functionalities simultaneously [65,73–75,78]. This is particularly beneficial for pavement evaluation systems. For instance, if the device is intended to perform complex and advance real-time analysis of pavement conditions, an SBC could be leveraged. SBC ability to handle complex tasks and perform real-time processing would be advantageous in scenarios where the device needs to quickly process large amounts of sensor data to detect potholes or cracks on the pavement in real-time. Moreover, if the device is expected to collect high-resolution images from pavement surfaces and execute complex algorithms such as machine learning for

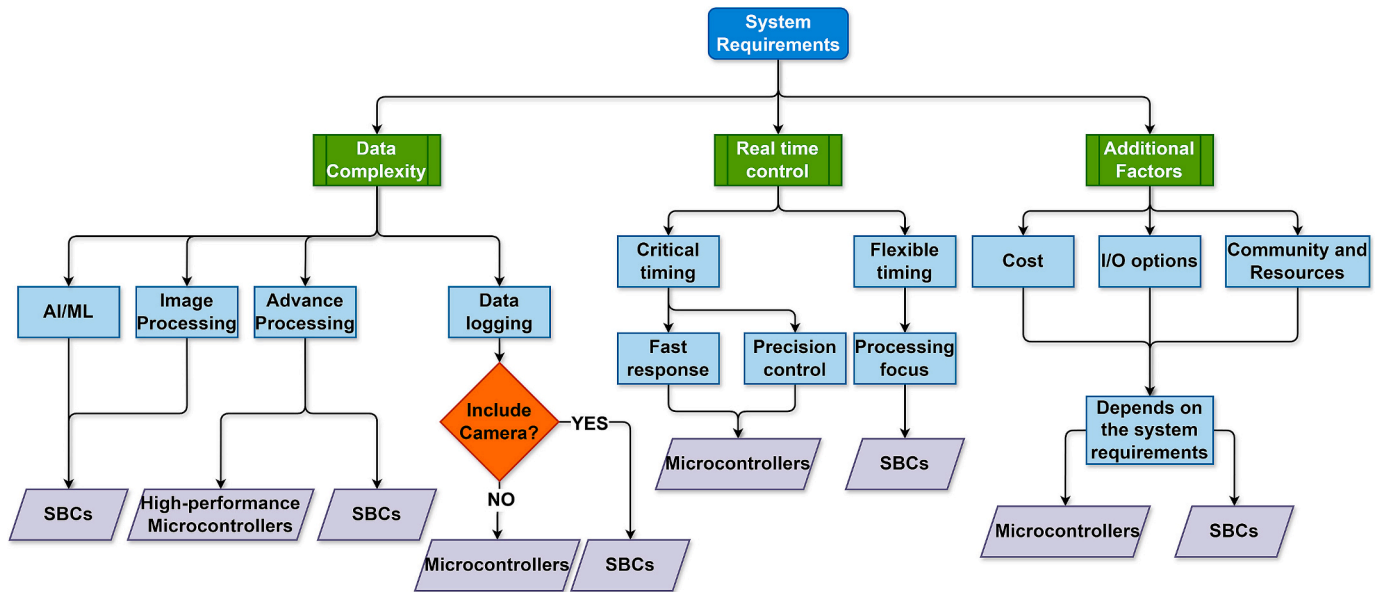


Fig. 1. Decision flowchart for selecting microcontrollers and SBCs in pavement evaluation systems.

Table 3

Examples of microcontroller platforms and their specifications.

Platform	Architecture	Specifications	Key Features
Arduino Uno Rev3	ATmega328P	16 MHz CPU, 2 KB RAM, 32 KB Flash	Ideal for beginners; simplicity, extensive community support [79,80].
ESP32-WROOM	Dual-core Xtensa LX6	Built-in WiFi/BLE, 240 MHz CPU, 520 KB RAM	Perfect for IoT projects with integrated wireless connectivity [81–83].
Raspberry Pi Pico (RP2040)	RP2040 (Dual-core ARM Cortex-M0+)	133 MHz CPU, 264 KB RAM, 2 MB Flash	Low-cost, flexible GPIO control, supports MicroPython and C/C++ [84,85].
STM32F746G-DISCO	ARM Cortex-M7	216 MHz CPU, 340 KB RAM, 16 MB SDRAM, 1 MB Flash	Designed for high-performance, real-time, and industrial-grade applications [83,86].

pavement distress detection, an SBC with advanced graphics processing would be an ideal choice. The high computational power of these SBCs allows them to efficiently process the high-resolution images and run sophisticated machine learning models. Additionally, SBCs with extensive connectivity options can seamlessly integrate and process data from multiple sensors simultaneously, including temperature sensors, distance sensors, and accelerometers. This enables the development of comprehensive pavement evaluation systems that leverage the advanced features and multitasking capabilities of these SBC platforms. Table 4 provides example options for common types of SBCs along with their specifications.

When comparing microcontroller options, the Arduino Uno is a popular choice due to its simplicity and ease of programming, particularly for beginners [79]. The ESP32, on the other hand, offers more processing power and advanced connectivity options, including built-in Wi-Fi and Bluetooth, making it suitable for applications that require wireless data transmission or integration with IoT networks [81–83]. The Raspberry Pi Pico is recognized for its impressive performance and versatility, enabling it to handle complex computational tasks efficiently while maintaining low power consumption [84]. The STM32 series of microcontrollers, developed by STMicroelectronics, are also widely used. These microcontrollers have high performance and excel at real-

Table 4

Examples of SBC platforms and their specifications.

Platform	Architecture	Specifications	Key Features
Raspberry Pi 4B*	Broadcom BCM2711 (Quad-core Cortex-A72)	1.5–2.0 GHz CPU, Up to 8GB RAM	USB 3.0, Gigabit Ethernet, 4 K video.
Orange Pi 5	RK3588S (Octa-core ARM)	2.4 GHz CPU, Up to 16GB RAM	PCIe, Multiple display outputs.
NVIDIA Jetson Orin Nano (DEV-22098)	6-core ARM Cortex-A78AE v8.2 + NVIDIA Ampere GPU	1.5 GHz CPU, 8GB	GPU acceleration, AI optimized.
NanoPi NEO4	Rockchip RK3399 (Hexa-core: Dual Cortex-A72 + Quad Cortex-A53)	1.5–2.0 GHz CPU, 1GB RAM	Compact design, USB 3.0, Gigabit Ethernet, 4 K video output, low power.

* This component is implemented in the case study system and detailed in subsequent sections.

time data acquisition and the execution of predefined algorithms [83]. The deterministic behavior and real-time response of STM32 microcontrollers ensure reliable and consistent performance in these types of applications.

When selecting an SBC for pavement evaluation systems, there are several prominent options to consider. The Raspberry Pi series is a popular choice due to its wide software ecosystem and features [87,88]. For applications requiring even greater processing power, the Orange Pi series may be the better fit, as it offers more advanced capabilities [89]. Another compelling SBC option is the NVIDIA Jetson Nano, which leverages specialized hardware to handle complex computational workloads efficiently [90], making it a strong contender for pavement evaluation systems that require advanced processing capabilities, such as AI and computer vision.

3.2.2. Cost

Microcontrollers generally offer a more low-cost solution for pavement monitoring applications. Their economic advantage stems from both lower hardware costs and simpler system requirements. The cost-effectiveness of microcontrollers extends beyond the initial hardware purchase - they require simpler power supplies, less complex supporting circuitry, and minimal additional components for basic operation. This

makes them particularly attractive for large-scale deployments or systems requiring multiple monitoring points.

SBCs, while more expensive than microcontrollers, offer greater functionality and processing power that may justify their higher cost in certain applications. However, the total system cost for SBCs must also account for additional requirements such as storage devices, power supplies, cooling systems, and enclosures. These supporting components can significantly increase the overall system cost compared to microcontroller-based solutions.

The relative costs of different platforms can be understood through examining their prices at the time of this study, though these values are subject to market fluctuations and availability. Table 5 presents a comparison of costs and associated requirements for different platforms.

3.2.3. Input/Output (I/O) options

The I/O capabilities of both microcontrollers and SBCs are crucial factors in pavement evaluation applications, as they determine the types and number of sensors that can be integrated into the monitoring system. Each platform category offers distinct I/O interfaces that influence their suitability for different sensing applications.

Microcontrollers provide diverse I/O options with direct hardware control capabilities [66,91,92]. The Arduino Uno Rev3 offers digital and analog I/O pins suitable for basic sensor integration, while the ESP32 extends these capabilities with additional GPIO pins and built-in wireless connectivity. The STM32 series provides advanced peripheral options including high-speed Analog-to-Digital Converters (ADCs) and multiple communication interfaces, making them suitable for precise sensor measurements. The Raspberry Pi Pico stands out with its programmable I/O functionality, allowing flexible interface configurations for various sensor types.

SBCs typically offer more sophisticated I/O options and standardized interfaces [66,93]. The Raspberry Pi 4B provides USB ports, HDMI output, and dedicated camera/display interfaces alongside GPIO pins. The Orange Pi 5 platforms expand these capabilities with additional interfaces like PCIe and multiple display outputs. NVIDIA Jetson boards include specialized I/O for camera inputs and display outputs, particularly beneficial for visual pavement inspection systems.

3.2.4. Real-time processing

Real-time processing capabilities are critical in pavement evaluation systems where timing and responsiveness are essential for accurate data collection and analysis. Microcontrollers and SBCs handle real-time processing in fundamentally different ways due to their architectural differences.

Microcontrollers excel in deterministic real-time operations due to their direct hardware control and lack of operating system overhead [94–97]. The STM32 series, for example, can provide precise timing control and immediate interrupt response, making them ideal for time-critical sensor sampling. ESP32's dual-core architecture allows dedicated real-time task handling while maintaining wireless communications.

SBCs approach real-time processing differently due to their operating system. While they offer more processing power, their standard Linux operating systems introduce timing uncertainties that can affect real-time performance [98,99]. Each platform's real-time capabilities influence their suitability for different pavement monitoring tasks. Microcontrollers are better suited for applications requiring precise timing

control and immediate sensor response. However, SBCs are preferable for applications where real-time requirements are less stringent but require complex data processing.

3.2.5. Community support and resources

The availability of development resources and community support significantly influences the implementation of pavement evaluation systems using these platforms. Both microcontrollers and SBCs offer varying levels of support and development resources, which can impact project development efficiency and long-term maintainability.

In the microcontroller category, Arduino platforms benefit from a broad developer community that has created an extensive collection of libraries and resources. The availability of expansion boards (shields) and the platform's ease of use make it particularly accessible for development projects [100,101]. The Arduino family's widespread adoption in sensor integration applications has made it one of the most well-documented and supported platforms for embedded systems development [100,101]. Among SBCs, the Raspberry Pi platform stands out for its extensive support ecosystem. It is widely used in IoT and sensor applications across technology companies, universities, and hobbyist communities. The platform's user-friendly nature and abundant resources make it particularly suitable for development projects [100,101].

4. Selecting sensors

A wide range of existing sensors can be connected to microcontroller and SBC boards and adapted for evaluating pavement condition indicators. While these sensors were not primarily designed for pavement-specific applications, they can be effectively utilized to measure various pavement condition indicators through innovative approaches. The following sections outline types of pavement indicators and the potential sensors that can be employed for this purpose.

4.1. Pavement roughness related sensors

Pavement roughness is a crucial parameter in evaluating road quality, impacting vehicle dynamics and ride comfort. It is defined as the deviation of a road surface from a true planar surface, with characteristic dimensions influencing vehicle behavior and ride quality [25,102,103]. The IRI is a widely accepted metric for quantifying pavement roughness, offering a standardized measure for assessing road conditions. There are three common methods for measuring pavement roughness:

- Using vehicle dynamics and signal processing: This approach estimates pavement roughness by analyzing the vertical acceleration or suspension displacement of a vehicle traversing the road surface. The vehicle's dynamic response to the road profile is used to infer the roughness characteristics [104].
- Direct elevation measurement: This method involves directly measuring the elevation profile of the road surface using specialized equipment, such as laser profilers [105]. The collected elevation data is then processed to calculate the roughness indices.
- Fusion of vehicle dynamics and elevation data: This approach combines the vehicle's dynamic response and direct elevation measurements to estimate pavement roughness. Inertial profilers are examples of systems that fuse these two data sources, providing a comprehensive assessment of road roughness [106].

Before choosing specific sensors for pavement roughness evaluation, several key criteria must be considered to ensure appropriate sensor selection. Fig. 2 shows different types of sensors that can be integrated and illustrates the main considerations in selecting sensors for pavement roughness measurement. The following sections detail different types of sensors that can be integrated into pavement roughness evaluation

Table 5

Cost Aspects of Microcontrollers and SBCs.

Platform Type	Cost (\$)
Entry-level Microcontrollers (Arduino Uno Rev3)	27
Advanced Microcontrollers (STM32F746G-DISCO)	57
Entry-level SBCs (Raspberry Pi 4B 8GB)	75
Advanced SBCs (NVIDIA Jetson Orin Nano 8GB)	500

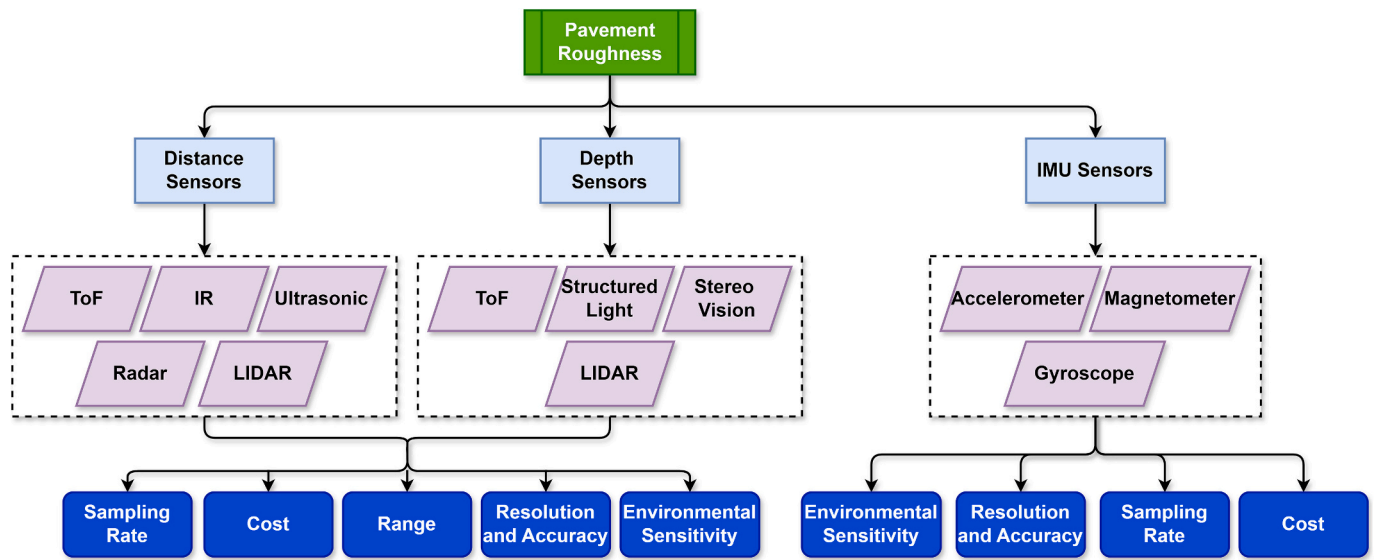


Fig. 2. Selection criteria and sensor categories for pavement roughness measurement for different measurement approaches.

systems, discussing their characteristics, capabilities, and limitations.

4.1.1. Distance measurement sensors

To measure the elevation profile of the pavement surface directly, various distance measurement sensors can be employed. These sensors can be broadly categorized into several main groups, including ultrasonic sensors, infrared (IR) sensors, LiDAR (Light Detection and Ranging) sensors, time-of-flight (ToF) sensors, and Radar sensors. Each sensor type possesses its own unique advantages and limitations, making them suitable for different applications and operating conditions. Table 6 presents examples of each sensor group, along with their respective advantages and disadvantages.

Ultrasonic sensors, also known as sonar sensors, are commonly used for distance detection [107,108]. These sensors operate by emitting high-frequency ultrasonic waves towards a target object and calculating the distance based on the wave's transit time and the speed of sound

[107–109]. They are not affected by the color, transparency, or brightness of the object being measured, although they may encounter challenges when measuring distances to objects with complex surfaces [107,108,110]. To improve accuracy, ultrasonic sensors are often combined with humidity and temperature sensors since the speed of sound is influenced by these factors [108,111]. On the other hand, IR sensors determine distance by emitting infrared light signals and analyzing the angle of reflection [108,110,112]. These sensors are sensitive to ambient light conditions and the surface color of the target object [108,110]. In contrast, LiDAR sensors merge aspects of IR and ultrasonic sensors, utilizing laser beams to measure distances with high precision [108,110,112,113]. LiDAR sensors excel in measuring distances to small objects with remarkable accuracy, although they can be costly and pose risks due to laser beams [108,110]. ToF sensors, falling under the LiDAR spectrum, estimate distances by measuring the time between pulse emission and reception using light pulses. By calculating the time taken for the light pulse to travel to and from the target object, ToF sensors determine distances accurately [108,110]. Radar sensors emit electromagnetic waves that can penetrate surface materials and operate effectively in various weather conditions including rain, snow, and fog [114–116]. Unlike optical sensors, radar sensors are not affected by lighting conditions or surface reflectivity, making them particularly useful for continuous monitoring applications. For example, pulsed coherent radar technology provides high-resolution measurements while maintaining a compact form factor, though it still requires sophisticated signal processing algorithms for optimal performance [117].

When selecting distance measurement sensors for pavement roughness evaluation, several critical factors must be considered to ensure accurate and reliable data acquisition. One crucial aspect is the sensor's range, which should be determined based on the intended mounting location relative to the pavement surface. For instance, if the sensor is mounted far from the pavement on a moving vehicle, a longer-range sensor may be required, whereas a shorter-range sensor could suffice for a setup closer to the pavement surface. The accuracy of distance measurements can be influenced by the height between the pavement surface and the sensor. As the distance increases, the potential for errors and noise in the measurements may also increase, particularly for sensors with lower resolutions or limited ranges.

Another important factor is the sampling rate, as it determines the sensor's ability to capture high-frequency vibrations and surface irregularities accurately. Sensors with higher sampling rates are generally preferred for pavement roughness evaluation to ensure that subtle details are not missed. Environmental conditions and ambient factors can

Table 6
Distance measurement sensor groups for pavement elevation profiling.

Sensor Group	Examples	Advantages	Disadvantages
Ultrasonic Sensors	HC-SR04	Inexpensive, easy to integrate, suitable for short-range measurements	Accuracy and performance affected by environmental factors (temperature, humidity), limited range
Infrared (IR) Sensors	Sharp GP2Y0A21YK	High resolution, measure the distance of complex surfaces	Limited effective range, influenced by ambient light conditions, low frequency
LiDAR Sensors	LIDAR-Lite v3	High resolution, high accuracy, longer measurement ranges	More expensive, could be harmful to human eyes
Time-of-Flight Sensors	*VL5311x	High resolution, high accuracy, longer measurement ranges	Performance affected by ambient light conditions and surface characteristics
Radar Sensors	*Acconeer A121	Robustness against environmental factors, penetrates surface materials, high accuracy	Higher cost, complex signal processing required

* This component is implemented in the case study system and detailed in subsequent sections.

also significantly impact sensor performance. For example, ambient light can affect the accuracy of infrared and time-of-flight sensors, while ultrasonic sensors may be susceptible to vehicle engine sound noise, necessitating the application of filters or signal processing techniques to mitigate these effects. Furthermore, considerations such as power consumption, communication interfaces, and integration with microcontrollers or SBCs should also be evaluated. Sensors with low power requirements and compatibility with common communication protocols like I2C, SPI, or UART can simplify the system design and extend the operational time of battery-powered or mobile data acquisition setups.

4.1.2. IMU sensors

IMU sensors, such as accelerometers, gyroscopes, and magnetometers, can be effectively utilized to assess pavement roughness. These sensors provide valuable data about the vibration, motion, and orientation of a vehicle traversing the road surface, enabling the estimation of pavement roughness through advanced signal processing techniques [51,118].

Accelerometers measure the acceleration forces experienced by an object in one or more axes [108,119]. When mounted on a vehicle, accelerometers can detect the vertical acceleration caused by the vehicle's interaction with the road surface. The vertical acceleration data can be processed to estimate the pavement roughness profile, as rougher surfaces result in higher acceleration variations. Different types of accelerometers, such as piezoelectric, piezoresistive, or Micro-Electro-Mechanical Systems (MEMS) accelerometers [120], can be employed depending on the desired accuracy, measurement range, and cost considerations.

For SBC and microcontroller boards, the most commonly used type of accelerometer is the MEMS accelerometer due to their compact size, high sensitivity, low power consumption, and reliability [121]. When choosing a MEMS accelerometer for the purpose of evaluating pavement roughness, factors such as the communication interface of the accelerometer with the board (ex. SPI or I2C), operating temperature, power requirements, sensitivity, and range must be taken into account. Additionally, the sampling frequency and noise density of the accelerometer should align with the specific requirements of the application and desired accuracy. In applications like vibration analysis when evaluating pavement roughness, MEMS accelerometers with low noise density and high sampling frequency are preferred for accurate data acquisition. A minimum range of ± 2 is required to have accurate measurement [122]. Table 7 shows examples of MEMS accelerometers that can be connected to microcontroller and SBC boards, along with their specifications.

While accelerometers are the primary sensors for capturing the essential data required for pavement roughness estimation, other motion-sensing sensors such as gyroscope can provide supplementary information to improve the accuracy and reliability of the measurements. Gyroscopes are sensors that measure the angular velocity or rotational rate of an object [123]. When combined with accelerometer data, gyroscope measurements can provide information about the vehicle's orientation and angular motion. This information is particularly useful for compensating for the effects of vehicle pitch and roll during pavement roughness evaluation, ensuring accurate measurements regardless of the vehicle's dynamic behavior.

Similar to accelerometers, the selection of gyroscope sensors for

pavement roughness evaluation systems involves considering various factors to ensure optimal performance and accurate measurements. However, instead of having separate accelerometer and gyroscope sensors, IMU sensors can provide both functionalities in a single package. IMU sensors typically integrate a three-axis accelerometer and a three-axis gyroscope, along with additional sensors like magnetometers, into a compact unit [123]. When selecting an IMU sensor, factors such as measurement range and sensitivity for both accelerometer and gyroscope components, noise performance and bias stability of the integrated sensors, operating temperature range, power consumption, communication interfaces, size and weight constraints, cost and availability, as well as integration requirements for combining accelerometer and gyroscope data should be carefully evaluated.

4.1.3. Depth camera sensors

Depth cameras, also known as RGB-D cameras or 3D cameras, are devices that can capture depth information in addition to regular color (RGB) images [124]. These cameras can provide valuable data for pavement roughness evaluation by generating detailed 3D models or point clouds of the road surface. Ultimately, the point cloud can be utilized to measure IRI and surface characteristics [125,126]. There are several types of depth cameras that can be considered for this application based on their working principles: Structured Light Cameras, ToF Cameras, Stereo Vision Cameras, and LiDAR-based Depth Cameras. Table 8 summarizes the key features and specifications of different types of depth cameras.

Selecting a depth camera for pavement roughness evaluation involves several key considerations. Depth accuracy and spatial resolution are crucial for precise quantification of pavement irregularities. These parameters are influenced by factors such as sensor noise, calibration, and environmental conditions, and are determined by the camera's imaging sensor and optics [127–130]. The operating range of the depth camera is another critical factor, as evaluations may need to be conducted at varying distances from the camera. ToF cameras and LiDAR-based sensors generally have longer operating ranges compared to structured light or stereo vision approaches [108,110], making them more suitable for applications requiring data acquisition from greater distances.

Furthermore, the frame rate and data throughput of the depth camera should be evaluated, especially for applications involving data capture while in motion. Adequate frame rates and data throughput are necessary to ensure sufficient sampling of the pavement surface geometry, even at higher velocities, to accurately represent surface irregularities. Environmental factors such as ambient lighting conditions, temperature, and vibrations can significantly influence the performance of depth cameras. For instance, structured light cameras may struggle in direct sunlight due to interference with the projected pattern, while ToF cameras and LiDAR are generally more robust to varying illumination conditions. Temperature and vibration can also affect sensor performance and calibration, necessitating careful consideration of the intended operating environment. Lastly, integration with SBC and microcontroller boards, is another important aspect to consider. While some depth cameras are designed for seamless integration with these platforms, others may require the installation of additional drivers or software libraries. Compatibility and potential limitations should be

Table 7
Examples of MEMS accelerometers and their specifications.

Model	Maximum Measurement Range (g)	Sampling Rate (Hz)	Typical Noise Density ($\mu\text{g}/\sqrt{\text{Hz}}$)	Communication Interface	Supply Voltage (V)	Power Consumption (μA)
ADXL345	± 16	10–3200	290, 430	SPI, I2C	2–3.6	0.1–140
LIS3DH	± 16	1–5300	220	SPI, I2C	1.7–3.6	0.5–185
MMA8451	± 8	1.56–800	99	I2C	1.95–3.6	1.8–165
MSA311	± 16	1–1000	300	I2C	1.62–3.6	1–130
H3LIS331DL	± 400	0.5–1000	1500	SPI, I2C	2.16–3.6	1–370

Table 8

Comparison of different depth camera types based on key features and specifications.

Depth Camera Type	Examples	Depth Sensing Principle	Depth Resolution	Operating Range	Outdoor Performance	Typical Cost
Structured Light	Microsoft Kinect V1, Intel RealSense D415	Infrared pattern projection and deformation analysis	High	Short to medium	Limited due to ambient IR interference	Low to moderate
ToF	Azure Kinect DK, Arducam ToF Camera	Time-of-flight measurement of infrared light	Moderate to high	Short to medium	Good	Moderate to high
Stereo Vision	Intel RealSense D455	Disparity analysis between two camera views	Moderate	Short to long	Good	Moderate
LiDAR-based	Ouster Ultra-Wide FOV OS0 32 Lidar, Velodyne Puck	Scanning laser rangefinder combined with RGB camera	High	Long	Excellent	High

thoroughly evaluated to ensure successful integration and data acquisition. Processing capabilities must also be considered in this integration - while basic processing can be performed on SBCs in real-time for simple applications, comprehensive 3D reconstruction and IRI calculation typically require more powerful computing resources or offline processing, particularly when operating at normal traffic speeds.

4.2. Pavement distresses related sensors

Pavement surface distresses, such as cracks, potholes, rutting, and raveling, are common indicators of deterioration in road infrastructure. Accurate detection and characterization of these distresses are crucial for effective maintenance and rehabilitation strategies, ensuring safe and efficient transportation networks. Various sensor technologies can be employed to evaluate pavement surface distresses, each with its unique capabilities and limitations. Fig. 3 presents the main sensor categories for pavement distress evaluation along with their key selection criteria. The following sections discuss in detail the characteristics and selection criteria of RGB cameras, thermal cameras, and IMU sensors for pavement distress detection and evaluation.

These sensors do not directly measure pavement distresses, but rather collect raw data (such as images or thermal signatures) that can be processed and analyzed to identify and characterize various types of pavement distresses. The raw sensor data must undergo image processing, pattern recognition, and/or machine learning analysis to extract meaningful information about pavement conditions. This processing step is crucial for converting the sensor data into actionable information about pavement distresses such as cracking, raveling, or potholes.

4.2.1. RGB camera sensors

RGB cameras are devices that capture images using the red, green,

and blue color model. They are widely used in various applications, including pavement distress detection when combined with SBC and microcontroller boards. This involves capturing images of the pavement with the RGB camera and then using image processing techniques and computer vision models to identify any distress. Several camera options exist for microcomputer boards, each with its own advantages and limitations. Table 9 provides a comparison of different camera models suitable for use with SBC and microcontroller boards.

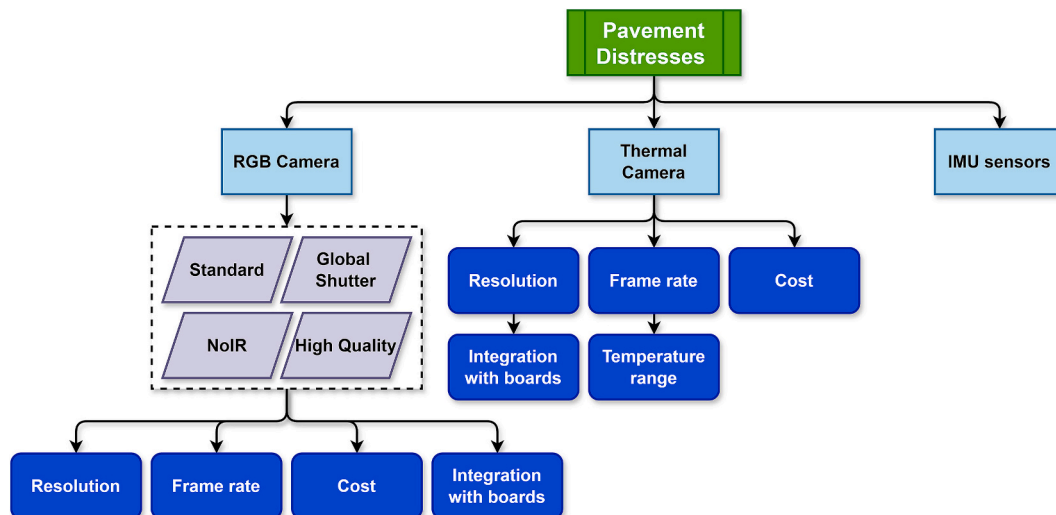
When selecting an RGB camera for pavement distress detection with microcomputers or microcontrollers, several key factors need to be considered. First, resolution is a crucial factor, as high-resolution cameras capture detailed textures of pavement surfaces, facilitating the identification of distress types like cracks and potholes. However, they

Table 9

Comparison of camera options for microcontrollers and SBC platforms.

Model	Example	Advantages	Disadvantages
Standard	*Raspberry Pi Camera Module V2	Low-cost and widely available	Distortion is possible with movement
High Quality	Raspberry Pi High Quality Camera	Higher resolution and captures more detail	More expensive/requires more processing power and storage
NoIR	Raspberry Pi NoIR Camera Module V2	Captures images in the dark	Not suitable for regular daytime use
Global Shutter	Raspberry Pi Global Shutter Camera	less distortion with movement	More expensive/lower resolution

* This component is implemented in the case study system and detailed in subsequent sections.

**Fig. 3.** Selection criteria and sensor categories for the detection of pavement distresses.

demand more processing power and storage, potentially constraining their use with resource-limited devices. Additionally, the sensor readout mechanism is important to consider. Rolling shutter cameras, prevalent in consumer-grade devices, scan the image sensor sequentially, potentially introducing distortion in images captured from a moving vehicle [131]. In contrast, global shutter cameras, which capture the entire image sensor simultaneously, minimize distortion and provide clearer images, making them ideal for applications involving movement [131].

Furthermore, the color gamut of the camera should be evaluated. For pavement distress detection, a standard RGB camera's color range is typically sufficient. Nevertheless, specialized cameras, such as NoIR cameras sensitive to infrared light, can offer benefits in specific scenarios, such as nighttime inspections. Moreover, the frame rate of the camera is another factor to consider. A moderate frame rate is usually sufficient for pavement distress detection with a moving camera, as very high frame rates can place significant processing demands on the microcomputer. Finally, cost and availability should be taken into account. Standard RGB cameras are cost-effective and readily available. However, applications requiring higher resolution, global shutter sensors, or specialized low-light capabilities may necessitate more expensive and less readily available cameras.

4.2.2. IMU sensors

In addition to RGB cameras, IMU sensors can be employed to aid in pavement surface detection and characterization. These inertial sensors, often integrated into microcontroller and SBC platforms, provide valuable motion and vibration data that can complement visual information from cameras. By mounting an accelerometer on a vehicle or mobile platform, the vertical accelerations experienced due to pavement roughness and irregularities can be measured. Abrupt changes in acceleration can indicate the presence of potholes, bumps, or other surface defects. Moreover, the magnitude and frequency of vertical accelerations can provide insights into the overall roughness and ride quality of the pavement surface.

The selection criteria for choosing appropriate accelerometer types, such as resolution, range, and sampling rate, were previously discussed in the context of motion-sensing sensor selection for pavement roughness evaluation. These factors remain relevant when selecting accelerometers and for pavement surface detection, as they influence the accuracy and reliability of the inertial sensor data.

4.2.3. Thermal camera sensors

Thermal camera sensors can provide valuable information about pavement surface conditions by detecting temperature variations that may indicate subsurface defects or structural issues [132]. These sensors, when integrated with microcontroller or SBC platforms, can capture thermal images or temperature measurements of the pavement surface. Temperature differentials can reveal delamination, moisture infiltration, or other hidden defects that might not be visible to RGB cameras. Table 10 examples of thermal camera sensors that can be integrated with microcontrollers and SBCs for pavement evaluation.

Table 10
Examples of thermal camera modules and their specifications and key features.

Module	Resolution	Temperature Range (°C)	Frame Rate (Hz)	Key Features
FLIR Lepton 3.5	160 × 120	−10 to 140	< 9.0	Small form factor, Low power, I2C/SPI interface
MLX90640	32 × 24	−40 to 300	64	Wide field of view, High refresh rate
AMG8833	8 × 8	0 to 80	10	Low cost, Easy integration
FLIR Boson	640 × 512	−40 to 550	60	High resolution, Advanced features

When selecting thermal sensors for pavement evaluation, several factors need consideration. Resolution is crucial for thermal imaging sensors, as higher resolution provides more detailed temperature distribution maps. Measurement range and accuracy are important factors, as pavement temperatures can vary significantly depending on environmental conditions. Additionally, sampling rate and response time affect the sensor's ability to capture temperature variations while moving. Integration complexity and cost must also be evaluated based on the specific requirements of the pavement monitoring system.

4.3. Pavement structural condition related sensors

Pavement structural condition is a crucial aspect of infrastructure management, as it directly impacts the performance, safety, and longevity of roads and highways [133]. Assessing the structural integrity of pavements is essential for planning maintenance, rehabilitation, and reconstruction activities [133]. This section provides an overview of the load cells, a key technology used in pavement structural evaluation.

4.3.1. Load cells

Load cells are sensors designed to detect the force applied to them and convert this force into an electrical signal, allowing for precise measurement of weight or force in various applications [134]. These sensors typically utilize strain gauges or other technologies such as piezoelectric elements to detect and quantify the force applied to them [56]. Load cells are widely used in pavement structural evaluation to measure surface deflections under simulated traffic loads. These sensors can provide valuable insights into the structural capacity and remaining service life of the pavement. When integrated with SBC and microcontroller platforms, they could offer a cost-effective and flexible solution for pavement condition assessment. Table 11 summarizes the characteristics of two commonly used load cell types in this context: strain gauge and piezoelectric load cells. These sensors offer different capabilities, making them suitable for various aspects of pavement assessment, from static load measurements to dynamic force analysis.

Load cell selection for microcontrollers and SBCs requires evaluating capacity, accuracy, environmental conditions, and budget. Additionally, the appropriate signal conditioning and amplification circuitry (such as the HX711 amplifier for strain gauge load cells) or analog-to-digital converters (ADCs) will need to be used to interface the load cell with some microcontroller or SBC platforms.

4.4. AI component integration for pavement evaluation

To enhance the capabilities of microcontroller and SBC platforms for pavement evaluation tasks, integrating specialized AI components can provide significant computational power and processing efficiency. This section explores key AI-powered hardware that can be directly

Table 11
Types of Load Cells Commonly Used with microcontrollers/SBCs for pavement structural evaluation.

Load Cell Type	Working Principle	Advantages	Disadvantages	Example Sensor
Strain Gauge Load Cells	Use strain gauges to measure deformation	High accuracy and precision	Temperature sensitive, requires signal amplification	TAL220
Piezoelectric Load Cells	Generate electric charge under applied force	Excellent for dynamic force measurements, high natural frequency	Not suitable for static measurements, charge leakage over time	EK1952

connected to boost their on-site analysis and processing abilities.

4.4.1. Edge AI accelerators

Edge AI accelerators are dedicated hardware that offload AI processing from the main CPU, bringing substantial computational power to small, portable devices. These accelerators are essential for applications requiring real-time processing and low power consumption, as they are tailored to meet the specific demands of edge computing scenarios, which often involve limited computational resources and strict latency requirements [135–137]. Table 12 presents examples of commercially available edge AI accelerators that can be integrated with microcontroller and SBC platforms.

The selection of an appropriate edge AI accelerator for pavement evaluation applications involves several critical factors. Computational performance is a primary consideration, as the accelerator must provide sufficient processing power to efficiently execute complex machine learning models for tasks such as image recognition, defect detection, and predictive analytics. Power efficiency represents another crucial factor, particularly for field deployments where energy consumption must be minimized to enable extended operation periods.

Platform compatibility serves as a fundamental requirement, ensuring seamless integration with the target microcontroller or SBC platform. Development ecosystem accessibility, including well-documented SDKs and development tools, significantly impacts the ease of integration and deployment of custom AI models. Cost-effectiveness must also be evaluated, balancing performance capabilities against project budget constraints.

4.4.2. AI sensors modules

AI sensor modules represent an emerging category of intelligent sensing devices that integrate both sensor hardware and AI processing capabilities into a single unit. These modules offer significant advantages for pavement evaluation systems by providing direct data acquisition and processing capabilities without requiring additional computational hardware. Table 13 presents examples of commercially available AI sensor modules that can be compatible with microcontroller and SBC platforms.

The selection of appropriate AI sensor modules for pavement evaluation applications requires consideration of several critical factors. Sensor resolution and quality significantly impact the accuracy of pavement defect detection and analysis capabilities. Processing speed becomes crucial for real-time applications, determining the module's ability to analyze pavement conditions during mobile inspections.

Integration capabilities represent another essential consideration, as the module must effectively communicate with the host platform while maintaining reliable data transmission. Power consumption characteristics particularly influence mobile and battery-powered deployment scenarios. Environmental resistance proves vital for outdoor pavement monitoring applications, where sensors must operate reliably under various weather conditions.

4.5. Positioning and navigation modules

Positioning and navigation modules are essential components in pavement evaluation systems, enabling precise location tracking, mapping of defects, and spatial data collection. These modules encompass

Table 12

Examples of Edge AI Accelerators for Microcontroller and SBC Platforms.

Accelerator Example	Processing Unit
Google Coral USB	Edge TPU
Intel Neural Compute Stick 2	Myriad X VPU
Sipeed MAix BiT	RISC-V K210
Grove AI HAT	Kendryte K210
Raspberry Pi AI HAT+	Hailo-8 or Hailo-8 L

Table 13

Examples of AI Sensor Modules for Microcontroller and Single-board computer Platforms.

Module	Processor and Camera Sensor	Built-in AI Functions
Raspberry Pi AI Camera	Sony IMX500 Image Sensor (12.3 MP)	Onboard AI processing for edge AI solutions [138].
HuskyLens	Kendryte K210, OV2640 (2 MP)	Face recognition, object tracking, object recognition, line tracking, color recognition, tag recognition, object classification [139].
Grove - Vision AI Module	Himax HX6537-A, OV2640 (2 MP)	Pre-installed ML algorithms for face and people detection; customizable models via Edge Impulse [140].
OpenMV Cam H7 R1	STM32H743VI, OV7725 (0.3 MP)	Color tracking, face detection, eye tracking, person detection, QR code detection, optical flow, and more [141].

GPS receivers, Global Navigation Satellite Systems (GNSS), and complementary positioning sensors. Table 14 presents examples of positioning modules compatible with microcontroller and SBC platforms.

The selection of appropriate positioning modules for pavement evaluation applications requires consideration of several critical factors. Positioning accuracy significantly impacts the precision of defect mapping and spatial data collection. Update rate determines the system's ability to track location changes during mobile inspections. Signal reception quality affects the module's reliability in various environmental conditions, particularly in urban areas with potential signal obstruction.

GPS accuracy can be limited in urban environments due to signal multipath effects, satellite visibility obstructions, and interference [142–144]. Several techniques can improve positioning accuracy in these challenging conditions. Real-Time Kinematic (RTK) GPS systems, like the ZED-F9P module, can achieve centimeter-level accuracy by using correction data from base stations. Sensor fusion with complementary technologies—combining GPS data with IMU measurements through Kalman filtering—can maintain accurate positioning during brief GPS signal loss [145]. Dead reckoning algorithms utilize vehicle speed and heading information to estimate position changes when satellite signals are compromised [146]. Additionally, enhanced receiver sensitivity and multi-constellation support significantly improve performance in urban canyons by increasing the number of available satellites.

Integration capabilities with the host microcontroller and SBC platform plays a crucial role in data acquisition and processing workflows. Power efficiency becomes essential for battery-powered deployment scenarios. Additional sensor integration, such as IMU or magnetometer capabilities, can enhance positioning accuracy and provide complementary motion data.

5. Case study: development of multi-sensor pavement evaluation system

The practical application of the selection criteria discussed in previous sections is demonstrated through a case study of a pavement evaluation system development. The system was designed to collect comprehensive pavement condition data, including roughness measurements and surface distress images, enabling detailed pavement condition assessment. The case study presented in this section is intended to demonstrate the practical application of the selection criteria and framework discussed in previous sections, not as a comprehensive validation of pavement condition algorithms. The primary objective is to illustrate how transportation agencies can develop their own customized devices based on specific requirements and constraints. The system shown here represents a proof-of-concept implementation that successfully collects raw sensor data which could be further processed using

Table 14

Examples of Positioning and Navigation Modules for Microcontroller and SBC Platforms.

Module	Key Features	Max Update Rate (Hz)	Horizontal, Velocity, and Heading Max Accuracy	Receiver Type
NEO-M8U GPS	Built-in Inertial Sensors, UDR Technology, High Sensitivity	30	2.5 m, 0.5 m/s, 1 degree	72 Channels GNSS Receiver
NEO-6 M GPS	GPS Only, Basic Positioning	5	2.5 m, 0.1 m/s, 0.5 degree	50 Channels, GPS L1 frequency
ZED-F9P-02B RTK	RTK Support, Centimeter-Level Accuracy	25	0.01 m, 0.05 m/s, 0.3 degree	multi-band GNSS and multi-band RTK technology
Ultimate GPS - PA1616S	Wide Voltage Range, Built-in Logging	10	< 3.0 m, 0.1 m/s	66 Channels, GPS L1 frequency
ZOE-M8Q	Ultra-Compact Size, Low Power Consumption, Multi-GNSS Support	18	2.5 m, 0.05 m/s, 0.3 degree	72 Channels GNSS Receiver

appropriate algorithms according to agency-specific needs. Detailed algorithm development, validation metrics, and performance analysis for specific condition indicators will be addressed in future research, as they would significantly expand the scope beyond the framework introduction intended in this paper.

5.1. System overview

The developed system integrates multiple sensors with a central processing unit to enable simultaneous collection of various pavement condition indicators. A Raspberry Pi 4B SBC serves as the main processing platform, chosen for its computational capabilities and extensive I/O options that facilitate integration with multiple sensors. Table 15 presents the cost breakdown of the system components and the purpose of selecting each component. The total system cost is approximately \$214. This cost structure demonstrates the feasibility of developing a comprehensive pavement evaluation system at a fraction of the cost of commercial solutions, which typically range from several thousand to tens of thousands of dollars.

5.2. Component selection rationale

The selection of each component was guided by specific requirements and the criteria outlined in previous sections. This component selection demonstrates the practical application of the criteria discussed, balancing performance requirements with cost considerations while ensuring system reliability and functionality.

5.2.1. Processing platform

The Raspberry Pi 4B (2GB version) was selected as the central

processing platform due to its balance of computing power, I/O capabilities, and cost-effectiveness. The 2GB RAM version provides sufficient processing capability for data collection and real-time analysis while maintaining system cost-efficiency. While microcontrollers could offer advantages in power efficiency and real-time control, they were not selected for this application due to several limitations. The simultaneous handling of multiple sensors (IMU, GPS, cameras, and distance sensors) requires significant processing power and memory resources that exceed typical microcontroller capabilities. Additionally, the need to process and store image data from the camera module, along with the requirement for complex data processing algorithms, made an SBC platform more suitable. The Raspberry Pi 4B's ability to run a full operating system also simplifies sensor integration and data management tasks, which would be more challenging to implement on a microcontroller platform.

5.2.2. Profile measurement sensors

XM125 A121 Radar Sensor using 60 GHz radar technology is adopted for its high accuracy and resolution in profile elevation measurements. The sensor provides distance measurements with mm resolution, operating range up to 20 m. Its key advantage lies in its robustness against environmental factors and immunity to ambient light conditions. VL53L1X ToF sensor is selected for its high resolution and accuracy in distance measurements with up to $\pm 3\%$ accuracy at ranges up to 4 m. The sensor features programmable timing budgets from 20 ms to 1000 ms for optimization of accuracy versus speed, and supports measurement rates up to 50 Hz. While this sensor is affected by sunlight and ambient light, its inclusion enables comparison with radar measurements to determine optimal sensor selection for pavement elevation measurement.

5.2.3. Motion and position sensors

The 6-Axis IMU (LSM6DS3TR-C) is chosen for its cost-effectiveness and suitable measurement range for pavement roughness applications. The integration of both accelerometer and gyroscope capabilities enables compensation for vehicle dynamics in elevation profile measurements. The sensor integrates a 3-axis accelerometer with ± 16 g measurement range and a 3-axis gyroscope with ± 2000 dps range. It features a sampling rate up to 1.6 kHz, suitable for capturing high-frequency vehicle vibrations induced by pavement roughness. The sensor's digital output with I2C/SPI interfaces enables direct integration with the Raspberry Pi platform. The Ultimate GPS Module (PA1616S) is selected based on its balance of cost and performance, offering reasonable accuracy (< 3.0 m) with a 10 Hz update rate, sufficient for spatial referencing of collected data. The module provides NMEA data output at 10 Hz update rate, enabling sufficient spatial resolution at typical highway speeds. It features a built-in antenna and serial communication interface, simplifying integration while minimizing additional hardware requirements. The module's -165 dBm tracking sensitivity ensures reliable signal reception during data collection. An external active GPS antenna is integrated to enhance signal reception quality, particularly important for accurate position tracking during mobile pavement

Table 15

Cost Breakdown of System Components (at the time of this study).

Component Category	Item	Purpose	Cost (\$)
Processing Platform	Raspberry Pi 4B (2GB)	Main processing platform	45.00
Profile Sensors	Acconeer XM125 A121	Radar sensor for pavement elevation measurements	45.00
	VL53L1X ToF	Laser distance sensor for pavement elevation measurements	14.95
Motion and Position	LSM6DS3TR-C IMU	For acceleration and gyroscope data	9.95
	Ultimate GPS Module PA1616S	For geospatial referencing	29.95
	Active GPS Antenna	To enhance signal reception quality	10.99
Imaging	Raspberry Pi Camera V2	For pavement surface imagery	25.00
Storage	128GB SD Card	For data storage	13.33
Power Supply	Lithium Battery Pack	To power the system	20.00
Total	–	–	214.17

surveys.

5.2.4. Imaging system

The Raspberry Pi camera module 2 is adopted for its direct compatibility with the Raspberry Pi platform, cost-effectiveness, and suitable image quality for pavement distress capture. The camera features an 8MP Sony IMX219 sensor, capable of capturing images with resolution up to 3280×2464 pixels. This resolution is sufficient for detecting various pavement distresses such as cracks, potholes, and surface deterioration. The camera supports frame rates up to 30 frames per second in high definition mode, enabling image capture even when the system is mounted on a moving vehicle. The direct integration with Raspberry Pi through the dedicated CSI camera port ensures reliable high-speed data transfer compared to USB cameras, while also reducing system complexity. The camera module also provides programmable controls for exposure, white balance, and other parameters, allowing optimization of image capture for varying lighting conditions encountered during pavement surveys.

5.2.5. Data storage and power

128GB SD card provides adequate storage capacity for extended data collection sessions. The storage capacity was selected based on the data collection requirements, considering the high sampling rates of sensors and image capture. A 20,000 mAh rechargeable lithium battery pack ensures continuous power supply during mobile operation, supporting the 5 V/3 A requirements of the Raspberry Pi 4B.

5.3. System integration and architecture

Following the component selection and system overview, this section details the technical implementation of sensor integration, power distribution, and data communication protocols. The architecture ensures synchronized multi-sensor data collection while maintaining system reliability and operational efficiency.

5.3.1. Overview of system integration

The system power distribution is managed through a 5 V/3 A supply

from the 20,000 mAh battery pack to the Raspberry Pi 4B via USB Type-C connection. The VL53L1X ToF sensor is connected to the Raspberry Pi's I2C bus using GPIO pin 3 Serial Data Line (SDA) and GPIO pin 5 Serial Clock Line (SCL), with power supplied through the 3.3 V and ground pins. The LSM6DS3TR-C IMU sensor, operating at I2C address 0×29 , is cascaded with the VL53L1X through the latter's I2C interface, enabling both sensors to share the same I2C bus. The GPS module connects through a USB-to-serial converter using TX and RX pins, with power supplied through the converter's 3.3 V and ground connections. An external active GPS antenna connects to the module via a uFL connector and uFL-to-SMA adapter cable to enhance signal reception. The Acconeer XM125 A121 radar sensor interfaces with the Raspberry Pi through its built-in USB Type-C port using a USB-C to USB-A cable. The Raspberry Pi Camera Module V2 connects directly to the dedicated CSI (Camera Serial Interface) port on the Raspberry Pi board. These connection protocols were selected to minimize pin conflicts and optimize the utilization of available Raspberry Pi GPIO pins. Fig. 4 illustrates the complete system connectivity diagram showing all component connections and communication interfaces.

5.3.2. Data acquisition and communication

Each sensor operates at different sampling rates optimized for its specific function and the system's processing capabilities. While the LSM6DS3TR-C IMU sensor is capable of sampling rates up to 1.6 kHz, it was configured to collect 3-axis acceleration and gyroscope data at 250 Hz in our implementation due to the processing constraints of managing multiple concurrent sensor streams on the Raspberry Pi 4B platform. This sampling rate provides sufficient temporal resolution for pavement roughness assessment while maintaining system processing efficiency.

The system implements parallel data collection through a multi-threaded architecture, with each sensor operating at optimized sampling rates: the ToF and radar sensors operate at 50 Hz and 2 kHz respectively, while the GPS module provides position updates at 10 Hz. The Raspberry Pi Camera V2 was configured to capture still images at 10 Hz using the video port rather than video recording. This approach was chosen because still images provide better quality for pavement distress

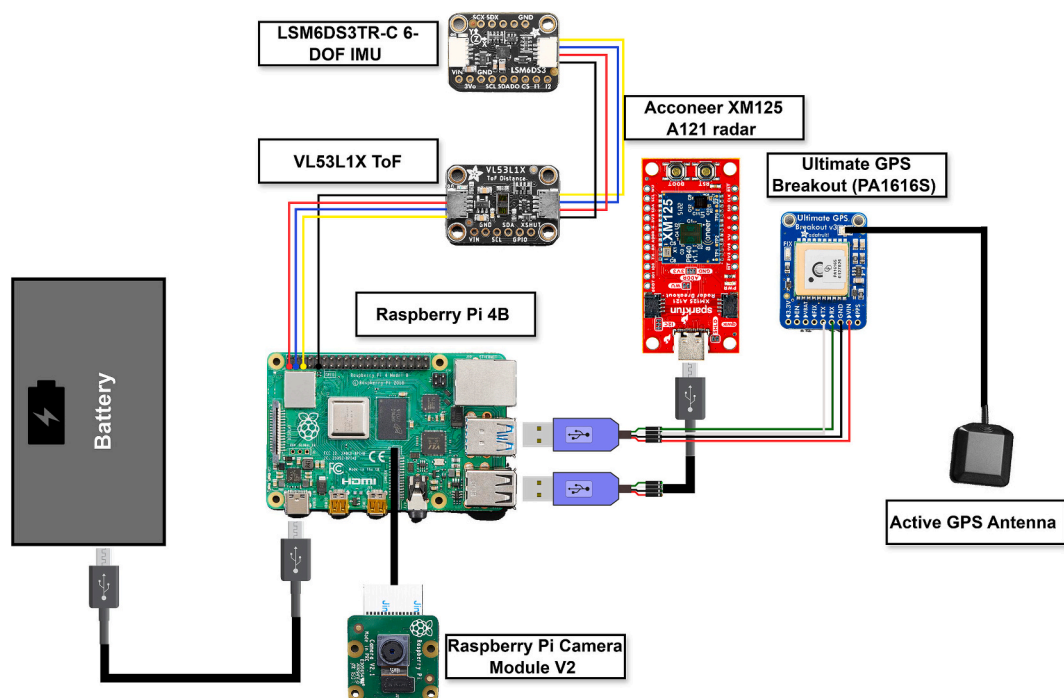


Fig. 4. System architecture and component connectivity diagram illustrating the integration of sensors, power supply, and communication interfaces with the Raspberry Pi 4B platform.

detection, while utilizing the video port (without actual video recording) enables faster image capture compared to traditional still capture methods.

A class threading architecture was implemented to enable efficient parallel processing and prevent data loss. The GPS class serves as the foundation for spatial referencing, continuously collecting GPS data and providing real-time geolocation information to other system components. This class maintains its own data stream while simultaneously making this information available to other threads for data tagging.

The Radar class manages continuous distance measurements from the radar sensor, with each measurement tagged with GPS coordinates obtained from the GPS class. This synchronization ensures that every distance measurement is properly georeferenced. The class handles data storage in CSV format, including fields for distance measurements, GPS coordinates, and timestamps, enabling comprehensive post-processing analysis.

The IMU class handles the collection of synchronized 3-axis

accelerometer and 3-axis gyroscope data. Each measurement set is tagged with GPS coordinates from the GPS class and includes precise timestamps. The data is stored in a separate CSV file that includes all IMU channels (three accelerometer axes and three gyroscope axes) along with their spatial and temporal references. This organization facilitates the analysis of vehicle dynamics and pavement response.

The Camera class manages the capture of pavement surface images at 1280×720 pixel resolution, maintaining a 10 Hz capture rate using the video port. During runtime, each captured image is automatically geotagged with GPS information from the GPS class. Images are stored with filenames formatted using timestamps, creating a clear temporal sequence of pavement surface conditions that can be easily correlated with other sensor data.

Two distinct data collection procedures were implemented to provide operational flexibility. The first method enables automatic data collection through a headless operation, utilizing the Raspbian OS Systemd service to initiate data collection immediately upon system power-

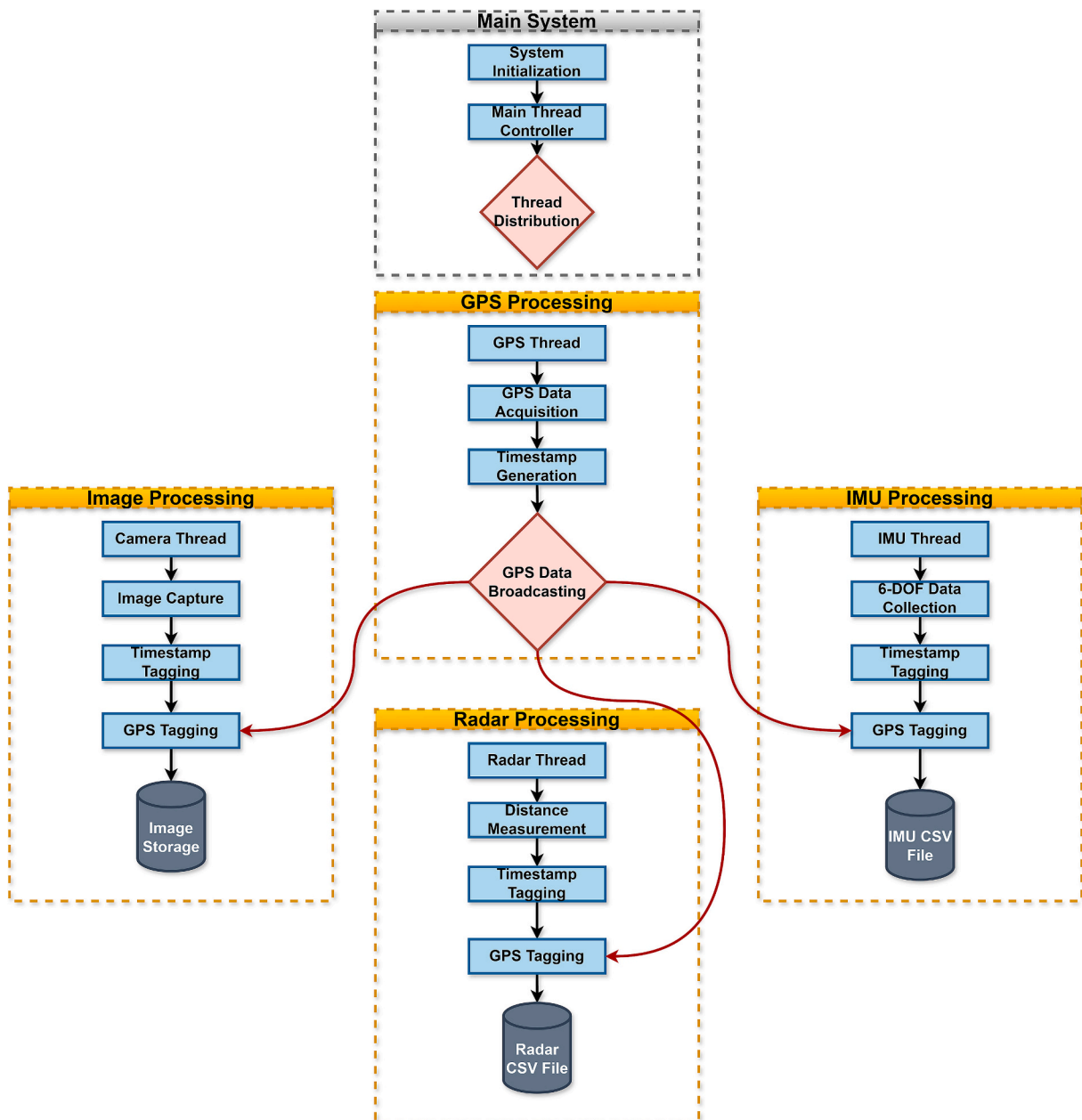


Fig. 5. Multithreaded system architecture for synchronized sensor data acquisition in the pavement evaluation system.

up. This approach enables autonomous operation without requiring user intervention. The second method implements remote operation capabilities through a WiFi connection, allowing system control via VNC connection from a smartphone or laptop. This provides full access to the Raspberry Pi desktop environment and enables real-time monitoring and control of the data collection process. Fig. 5 provides a comprehensive illustration of the multithreading architecture, demonstrating the relationships and data flow between different system components. The diagram shows how each sensor class operates independently while maintaining synchronized data collection through GPS integration and timestamp coordination.

5.3.3. Physical implementation

The system components are housed in a 12 × 12 in. box enclosure designed to protect against environmental factors while allowing proper sensor positioning, as shown in Fig. 6. The enclosure includes ventilation features to manage heat dissipation from the Raspberry Pi and other components and allow air flow.

The enclosure is mounted on an L-shaped steel beam structure that enables secure attachment to the vehicle, as shown in Fig. 7. The vertical beam attaches to the vehicle's hitch receiver, with a hitch fastener used to eliminate movement and ensure stability during operation. Two telescopic adjustment mechanisms are incorporated into the mounting system - one for height adjustment and another for horizontal offset. This adjustable mounting solution enables optimal positioning of the sensors relative to the pavement surface.

The sensors are mounted at specific positions within and outside the enclosure based on their functional requirements. The ToF sensor and camera are mounted with clear line of sight to the pavement surface through openings in the enclosure's bottom face. The radar sensor is mounted externally on the bottom of the enclosure to prevent interference with radar waves from the enclosure material. The IMU is rigidly mounted to capture vehicle dynamics accurately, while the GPS antenna is positioned outside the box for optimal satellite reception.

5.4. Sample system data collection

To demonstrate the functionality of the developed system, sample data was collected during preliminary field testing. Fig. 8 shows raw acceleration measurements collected by the IMU sensor along three axes (X, Y, Z) during vehicle movement over a pavement section. The accelerometer data demonstrates the system's ability to capture vehicle dynamic responses at the configured sampling rate of 250 Hz. Fig. 9 shows the angular velocity measurements from the gyroscope along three axes, which can be used to account for vehicle rotational movements during profile measurements.

Fig. 10 presents simultaneous distance measurements obtained from the XM125 A121 radar sensor, illustrating the system's capability to collect profile data simultaneously with IMU and GPS. Initial testing revealed significant limitations of the VL53L1X ToF sensor in outdoor conditions. Stationary testing revealed ± 2.5 mm accuracy in indoor controlled environments, but complete failure (0 % successful readings) in all outdoor lighting conditions tested, including both shadowed and direct sunlight environments, due to interference from ambient light. In stationary field testing, the radar sensor demonstrated excellent precision with a standard deviation of only 0.40 mm and accuracy exceeding 99 % in displacement detection mode, while the ToF sensor failed to provide any usable data under the same outdoor conditions. This performance difference highlights the importance of selecting sensors that are robust against environmental factors for outdoor pavement evaluation. The radar sensor proved to be a more suitable alternative, providing reliable measurements with millimeter resolution regardless of ambient lighting conditions.

To evaluate the system's data collection reliability, multiple runs were conducted on a 0.3-mile Portland Cement Concrete (PCC) pavement section. Table 16 presents the statistical analysis of four separate data collection runs, focusing on vertical acceleration (Z-axis) from the IMU and distance measurements from the radar sensor. These parameters were selected as they represent the primary inputs for pavement roughness calculation algorithms and profile assessment.

Analysis of measurement consistency across the four runs revealed excellent repeatability across all statistical measures. For the vertical

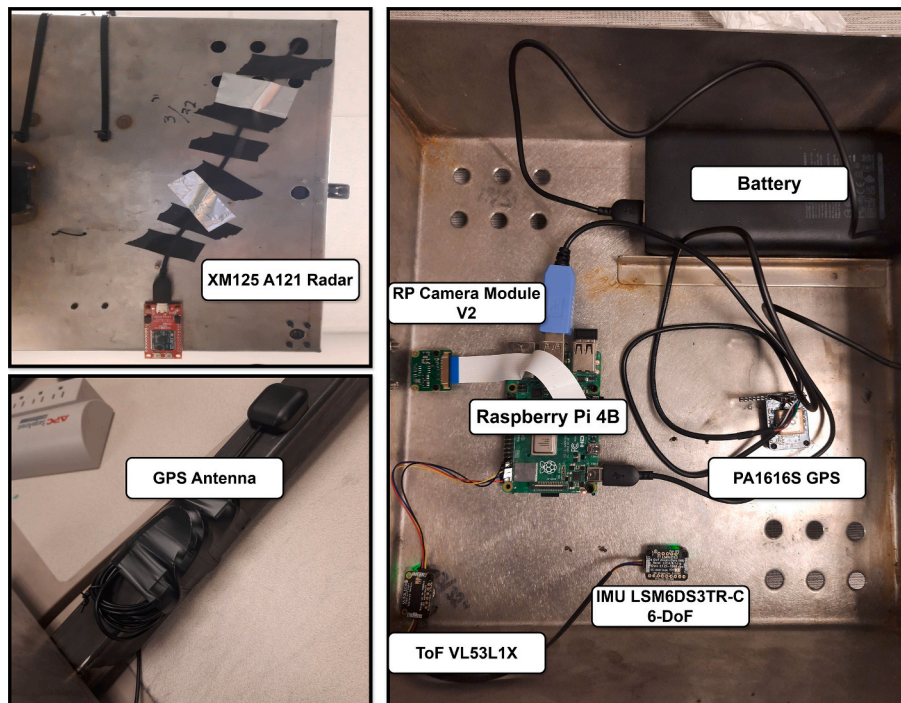


Fig. 6. System enclosure showing component layout and sensor positioning.



Fig. 7. Complete mounting system showing L-beam structure with telescopic adjustments and vehicle attachment mechanism.

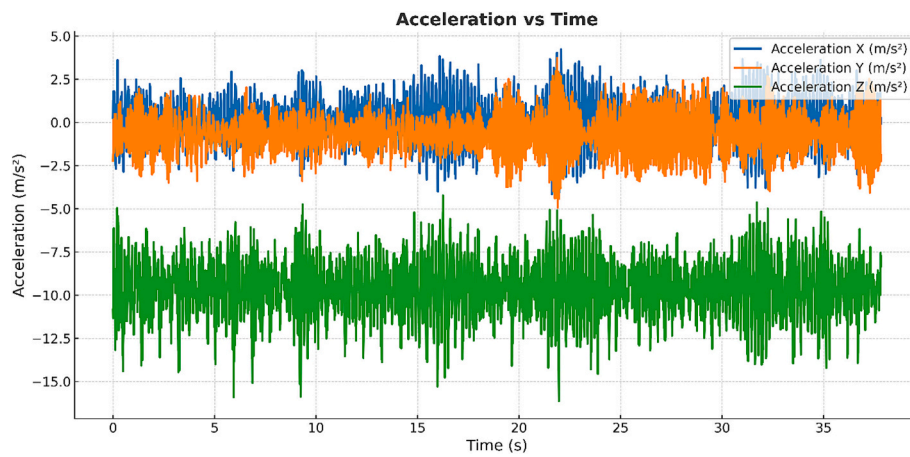


Fig. 8. Raw acceleration data from IMU sensor showing X, Y, and Z axes measurements collected at 250 Hz sampling rate.

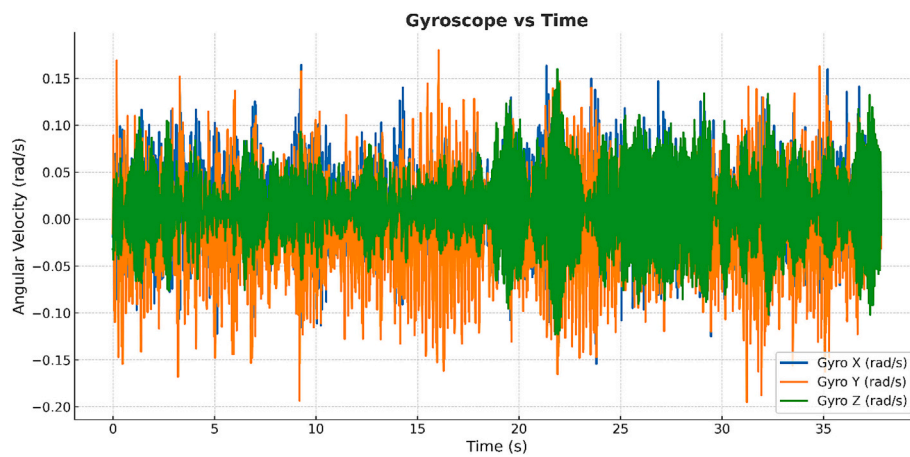


Fig. 9. Raw angular velocity data from IMU sensor showing X, Y, and Z axes measurements collected at 250 Hz sampling rate.

acceleration measurements, the coefficient of variation (CV%) between runs was remarkably low: 0.13 % for mean values, 0.21 % for median values, 0.20 % for Root Mean Square (RMS) values, 2.18 % for standard deviation, and 4.44 % for Power Spectral Density (PSD). The radar distance measurements demonstrated even greater consistency, with CV % values of 0.00 % for mean values (identical across runs), 0.0028 % for median values, 0.0001 % for RMS values, 2.16 % for standard deviation, and 3.06 % for PSD.

While the overall signal levels remained highly consistent between

runs (as evidenced by the low CV% values for mean and median), the standard deviation within each run (averaging 2.04 m/s^2 for acceleration and 9.44 mm for radar distance) reflects the system's sensitivity to detecting variations in the pavement surface. This combination of between-run consistency and within-run sensitivity is ideal for pavement condition assessment, as it ensures that variations detected are attributable to actual pavement characteristics rather than measurement instability.

The PSD analysis shown in Figs. 11 and Fig. 12 further illustrates the

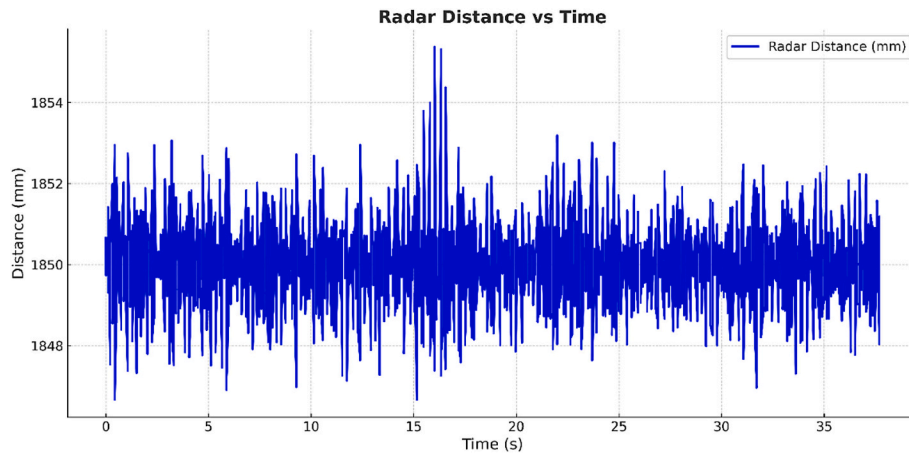


Fig. 10. Distance measurements from radar sensor showing pavement surface profile data collection capability.

Table 16

Statistical analysis of vertical acceleration and radar distance measurements across four repeated runs on a 0.3-mile PCC pavement section.

Data	Runs	Mean	Median	Std.	RMS	PSD
Acceleration Z (m/s ²)	1	-9.5794	-9.6096	2.0241	9.791	16.6485
	2	-9.5886	-9.5761	2.0527	9.8059	17.2826
	3	-9.5604	-9.5641	2.0161	9.7707	17.2110
	4	-9.5849	-9.594	2.1215	9.8169	18.5002
Radar Distance (mm)	1	1850.0000	1849.9245	9.416	1850.024	303.8933
	2	1850.0000	1850.0241	9.4571	1850.0242	293.6697
	3	1850.0000	1850.0295	9.2027	1850.0229	306.4744
	4	1850.0000	1850.0351	9.6995	1850.0254	316.2772

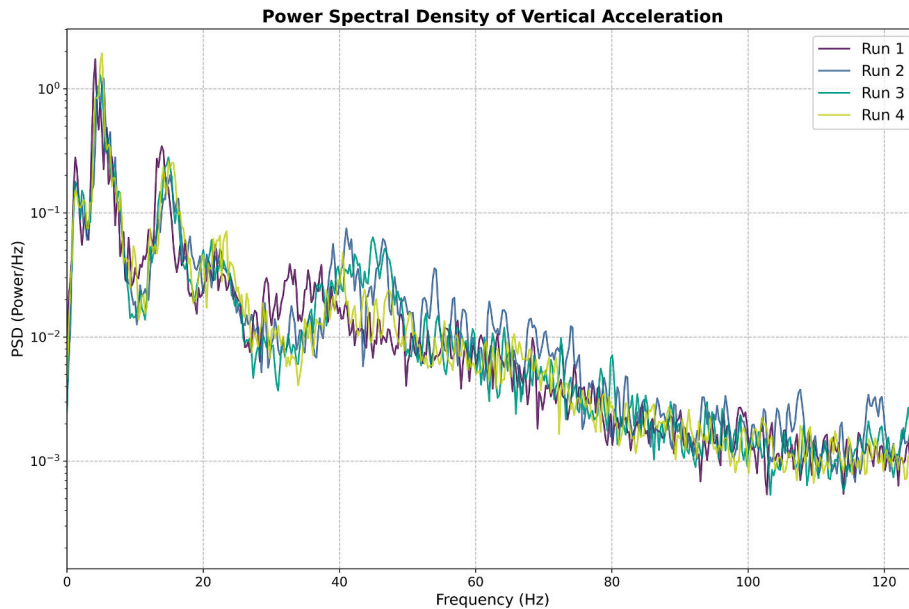


Fig. 11. PSD analysis of vertical acceleration data showing consistent frequency response patterns across four repeated measurement runs on the same pavement section.

frequency characteristics of both acceleration and distance measurements. Fig. 11 presents the PSD analysis of vertical acceleration data across all four runs, showing consistent frequency response patterns particularly in the lower frequency range (0–20 Hz) that typically corresponds to pavement irregularities. Similarly, Fig. 12 displays the PSD analysis of radar distance measurements, demonstrating remarkable consistency in frequency components across all runs. These PSD plots validate the repeatability of the measurement system by confirming that

the frequency signatures remain stable across multiple data collection passes. These statistical measures and spectral analyses confirm that the developed system can consistently capture pavement-induced vibrations and surface profiles with sufficient sensitivity and repeatability for subsequent condition assessment applications.

Fig. 13 provides example images captured by the system's camera, showing clear views of the pavement surface. The Raspberry Pi Camera V2 maintained consistent image quality across all test speeds, with a

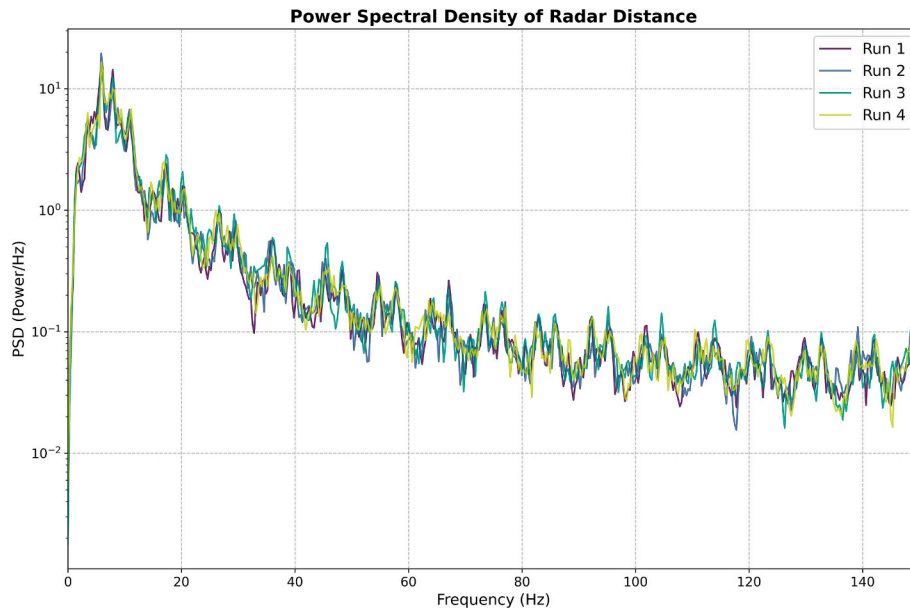


Fig. 12. PSD analysis of radar distance measurements demonstrating measurement consistency across four repeated runs on the same pavement section.



Fig. 13. Sample pavement surface images (1280×720 pixels) system's imaging capabilities.

sample RGB image resolution of 1280×720 pixels at 10 Hz capture rate. At the mounting height of 2.70 m, the camera achieved a spatial resolution of approximately 2.55 mm per pixel horizontally and 3.33 mm per pixel vertically, based on the camera's field of view (Horizontal FOV: 62.2° , Vertical FOV: 48.8°). At the lower mounting height of 1.85 m, the spatial resolution improved to approximately 1.74 mm per pixel horizontally and 2.28 mm per pixel vertically, offering enhanced detail for finer surface features. This resolution is sufficient for capturing surface texture and visible distresses such as cracks and raveling. At the configured resolution and frame rate, the system successfully captured and stored approximately 1200 spatially-referenced images per mile of roadway (at average speed of 30 mph), maintaining consistent image quality throughout the testing period. These initial results validate the system's capability to simultaneously collect and store multiple types of pavement condition data with proper spatial referencing.

5.5. System limitations and future work

The current implementation of the system revealed several important limitations during initial testing. These limitations can be categorized into hardware constraints, environmental factors, implementation challenges, and methodological considerations.

5.5.1. Hardware limitations

The ToF sensor proved unsuitable for outdoor pavement measurements due to its high sensitivity to sunlight and ambient light conditions, consistent with manufacturer specifications indicating optimal performance in controlled indoor environments. While the radar sensor provided more reliable measurements in outdoor settings, its absolute accuracy remains to be systematically validated against reference profiling systems. The system's processing capacity presents certain

constraints inherent to the Raspberry Pi platform when managing concurrent high-frequency data streams from multiple sensors, particularly when capturing and processing camera images simultaneously with IMU and radar data. These processing considerations would need to be addressed when implementing computationally intensive real-time analysis algorithms in future iterations of the system.

5.5.2. Environmental and operational constraints

Environmental factors significantly impact system performance. The ToF sensor's high sensitivity to ambient light conditions makes it unsuitable for outdoor pavement monitoring applications, as demonstrated during field testing where no reasonable measurements could be obtained in direct sunlight. The mounting system, while incorporating adjustable features, may require additional vibration isolation mechanisms at higher speeds to minimize measurement noise in both radar and IMU data. Vehicle-specific influences on IMU measurements present an additional consideration, as acceleration patterns can vary depending on the vehicle's suspension characteristics and mounting configuration, suggesting that vehicle-specific calibration approaches may be beneficial for optimal results across different vehicle types. The significant impact of environmental conditions on sensor performance underscores the importance of environmental robustness as a critical selection criterion when choosing sensors for pavement monitoring applications.

5.5.3. Data management and power constraints

Power management and storage capacity are important factors to consider for practical implementation. The 20,000 mAh battery supports around 11 h of continuous operation, providing sufficient capacity for typical data collection sessions while maintaining system portability. At an average survey speed of 30 mph, this translates to approximately 330 miles of pavement coverage on a single charge, making the system suitable for comprehensive network-level assessments without requiring frequent recharging. For extended surveys covering larger road networks, transportation agencies could implement power management strategies such as battery swapping or vehicle power integration.

The system generates approximately 1.2 GB of data per mile of pavement surveyed, including all sensor readings and captured images. With the 128 GB SD card, and accounting for approximately 8 GB required for the Raspberry Pi 4B operating system and software, the device can store data from approximately 100 miles of continuous pavement monitoring before requiring data offloading. These storage requirements align with typical data management practices of transportation agencies, though agencies implementing large-scale deployments would benefit from developing efficient data handling protocols to optimize storage utilization and streamline processing workflows.

5.5.4. Methodological enhancement opportunities

From an implementation perspective, the current proof-of-concept focuses primarily on data collection capabilities, with opportunities to develop automated analysis algorithms for converting raw sensor data into standardized pavement condition metrics in future work. The system's spatial referencing relies on standard GPS positioning, which provides adequate location tracking for segment-level condition assessment while offering potential for enhancement through differential GPS techniques when precise defect localization is required. Integration of the various sensor data streams is achieved through timestamp synchronization, which functions effectively at typical operating speeds while presenting opportunities for advanced synchronization techniques to further refine multi-sensor data correlation at higher vehicle velocities. These aspects represent natural evolution points as the system progresses from proof-of-concept to production-ready implementation.

5.5.5. Future research directions

Building on the insights from this implementation, several promising

research directions could further advance low-cost pavement monitoring systems. Potential processing enhancements through hardware or software optimization might enable more sophisticated real-time analysis while potentially reducing storage requirements. Further research could investigate methodologies for estimating standard pavement indices such as IRI using the types of sensor data collected by these systems, potentially combining IMU and profile measurements to calculate roughness metrics aligned with industry standards.

For systems incorporating visual data, computer vision and machine learning approaches present intriguing possibilities for automated distress detection, potentially enabling the classification of various pavement distresses such as cracks, potholes, and raveling from collected imagery. Positioning accuracy could potentially be improved through exploration of differential GPS or sensor fusion techniques to enhance spatial referencing of detected conditions.

The framework presented in this paper could be extended to accommodate additional sensing modalities in future research. Depth sensing technologies like 3D cameras could offer more detailed surface measurements, while thermal imaging might provide insights into sub-surface conditions not visible through conventional methods.

Researchers working in this domain might consider validation studies comparing measurements from low-cost systems against established pavement condition assessment methods. Such studies could evaluate performance across diverse pavement types, environmental conditions, and operational scenarios to establish the practical boundaries of these approaches. While this research demonstrates the development of a low-cost system (\$214 total cost), a comprehensive cost-benefit analysis comparing performance capabilities and limitations against standard commercial methods can be conducted in future work to evaluate the overall cost-effectiveness of this approach for different agency applications. This knowledge would be valuable for transportation agencies considering these technologies for their pavement management programs.

6. Conclusions

This paper presented a comprehensive framework for developing low-cost, embedded-based pavement evaluation systems using microcontroller and SBC platforms. The research establishes a set of selection criteria and integration guidelines that enable transportation agencies to create customized monitoring solutions tailored to their specific requirements and budgetary constraints. The case study demonstrates the successful application of the framework, featuring the development of a multi-sensor device capable of simultaneously collecting raw data from various sensors that can be processed to evaluate different pavement condition aspects, including geotagged profile measurements, vibration responses, and surface imagery.

The systematic selection criteria developed for microcontrollers and SBCs provide transportation agencies with a clear decision-making framework based on specific application requirements. These criteria—focused on purpose and complexity, cost considerations, I/O capabilities, real-time processing requirements, and community support—offer practical guidance for selecting the most appropriate platform for pavement monitoring applications. Similarly, the sensor selection guidelines across categories like distance measurement, motion sensing, and imaging systems enable agencies to match sensor characteristics with specific monitoring requirements, considering factors such as environmental robustness, measurement accuracy, and integration capabilities.

The research findings highlight the feasibility and affordability of developing embedded-based pavement monitoring systems using off-the-shelf components and open-source tools. The total system cost of approximately \$214 represents less than 1 % of the cost of commercial solutions, demonstrating a significant economic advantage for budget-constrained agencies. Quantitative analysis of system performance demonstrated excellent reliability, with statistical measures showing

coefficients of variation below 1 % for most metrics across repeated measurements. The system's 20,000 mAh battery supports approximately 330 miles of continuous pavement data collection at 30 mph on a single charge, while the 128 GB storage capacity accommodates approximately 100 miles of comprehensive multi-sensor data before requiring offloading.

The modular approach allows for flexibility in sensor selection and configuration, enabling agencies to prioritize data collection based on their unique needs. The selection criteria established in this research provide a systematic methodology for component evaluation, considering factors such as performance requirements, environmental conditions, and budget limitations. The integration of AI-based edge processing capabilities further enhances the potential for real-time analysis and decision support, leveraging the growing capabilities of compact computing platforms.

The implications of this research are significant for transportation agencies, particularly those with limited resources. By adopting the proposed framework, agencies can develop affordable, fit-for-purpose pavement evaluation systems that provide comprehensive condition data at a fraction of the cost of traditional methods. This empowers agencies to make data-driven maintenance decisions, optimize resource allocation, and improve overall pavement management practices.

The comparative sensor performance analysis provided valuable insights into environmental robustness requirements for pavement monitoring applications. While the radar sensor demonstrated excellent measurement stability with millimeter resolution in outdoor environments, the ToF sensor's susceptibility to ambient light highlights the importance of sensor selection based on operational conditions. These real-world observations validate the utility of our comprehensive selection criteria and demonstrate how agencies can leverage this framework to avoid costly trial-and-error approaches when developing their own systems.

In conclusion, this research provides a solid foundation for the development of low-cost, embedded-based pavement evaluation systems. The proposed framework, selection criteria, and case study demonstrate the potential for transportation agencies to create customized monitoring solutions that are both low-cost and technologically advanced. While further research into automated analysis techniques and comprehensive validation studies would enhance these systems' capabilities, the current framework already offers a practical pathway for agencies to implement basic pavement monitoring. By empowering agencies with affordable tools for comprehensive pavement condition assessment, this research contributes to the broader goal of improving infrastructure management practices and ensuring the longevity and safety of our transportation networks.

Funding sources statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of AI use

During the preparation of this work the author(s) used generative AI tools (Claude) for proofreading and improving the manuscript's language and readability. The author(s) reviewed and edited all AI-suggested content and take(s) full responsibility for the content of the published article.

CRedit authorship contribution statement

Yazan Ibrahim Alatoom: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Omar Smadi:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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